

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\131-C.pvdb

Date of Analysis : Apr 21,2026 9:40am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 131-C Page 2 of 161
 Vessel Design Summary: Step: 31 9:40am Apr 21,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2250.000 x 2256.000 mm ID
 Vessel Design Length, Tangent to Tangent 7163.00 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 275 °C
 Channel Side Design Temperature 250 °C
 Shell Side Design Pressure 4950.000 KPa.
 Channel Side Design Pressure 17000.002 KPa.
 Shop Shell Side Test Pressure 6481.927 KPa.
 Field Shell Side Test Pressure 6449.152 KPa.
 Shop Channel Side Test Pressure 22100.000 KPa.
 Field Channel Side Test Pressure 22286.064 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 TP316L	...	...	S31603	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel head	17000.002	101.30	17143.000	3.0000	N/A	No
Channel shell	17020.693	101.30	17183.000	3.0000	N/A	No
Shell flng.	4950.000	0.00	4925.000	3.0000	N/A	No
Shell	4969.796	0.00	5072.000	3.0000	N/A	No
Shell head	4969.797	0.00	5175.000	3.0000	No	No

Liquid Level: 2256.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 0.894

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Vessel Design Summary:

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Sphere	0.0	0.0	87.0	74.0	73.4	6.5	1.00	1.00
Cylinder	1500.0	1500.0	155.0	155.0	153.3	10.2	1.00	1.00
Body Flg	2213.0	370.0	340.0	300.0	294.6	194.5	1.00	1.00
Cylinder	7113.0	4900.0	46.0	46.0	44.9	...	1.00	1.00
Ellipse	7163.0	50.0	54.0	45.9	44.0	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Tube inlet	WRC-297	0.986	Shell	Passed
N1- inlet	WRC-297	0.986	Shell	Passed
Shell inlet-N3	WRC-297	0.964	Shell	Passed
Shell Outlet-N5	WRC-297	0.964	Shell	Passed
	WRC-107/537	0.856	n/a	Passed

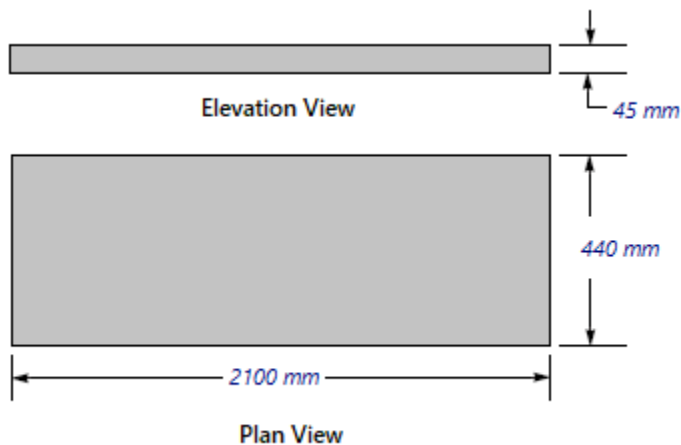
Foundation/Support Summary

Saddle Parameters:

Saddle Width	204.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1752.000 mm
Wear Pad Width	254.000 mm
Wear Pad Thickness	16.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1100.000 mm

Baseplate Length	2100.000 mm
Baseplate Thickness	45.000 mm
Baseplate Width	440.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	20.000 mm
Web Thickness	20.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	8

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	143156.19 Kg
Maximum Transverse Saddle Shear Load	3365.17 Kg



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Vessel Design Summary: Step: 31 9:40am Apr 21,2026

Maximum Longitudinal Saddle Shear Load	6730.34	Kgf
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Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	20883.67	Kgf
Maximum Transverse Saddle Shear Load	247.19	Kgf
Maximum Longitudinal Saddle Shear Load	315.48	Kgf

Weights:

Fabricated - Bare w/o removable internals	89033.8	kg.
Shop Test - Fabricated + water (full)	120726.5	kg.
Shipping - Fab. + rem. intls.+ shipping app.	89033.8	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	92610.9	kg.
Empty - Fab. + intls. + details + wghts.	92610.9	kg.
Operating - Empty + operating liquid (no ca)	116333.6	kg.
Field Test - Empty weight + test medium (full)	122953.6	kg.

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Nozzle Summary:

Step: 29 9:40am Apr 21,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Manhole	17143	OK	...	OK	44.98	OK	Passed
N1- Tube inlet	17183	OK	...	OK	20.97	OK	Passed
N1- inlet	17183	OK	...	OK	20.97	OK	Passed
Shell inlet-N3	5072	...	...	OK	17.01	OK	Passed
Shell Outlet- N	5072	...	...	OK	17.01	OK	Passed
N5	5175	...	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5072 Nozzle : Shell inlet-N3 [Shellside]
 Minimum MAWP Nozzles : 17143 Nozzle : Manhole [Tubeside]

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	Manhole	0.000	0.000	1231.200
20	N1- Tube inlet	750.000	0.000	627.850
20	N1- inlet	750.000	180.000	627.850
40	Shell inlet-N3	4713.000	0.000	1396.000
40	Shell Outlet- N3	4713.000	180.000	1436.000
50	N5	0.000	0.000	662.850

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 28 9:40am Apr 21,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1- Tube inlet	12.000 in	160	WNF	12.750	33.325	...	...	699.69	...
N1- inlet	12.000 in	160	WNF	12.750	33.325	...	...	699.69	1500
N5	16.000 in	160	None	16.000	40.488	...	...	561.41	...
Manhole	24.000 in	Actual	WNF	27.701	47.000	...	...	693.56	1500
Shell inlet-N3	30.000 in	Actual	None	30.000	35.000	...	...	412.29	...
Shell Outlet- N3	30.000 in	Actual	None	30.000	25.000	...	...	412.29	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1- Tube in	SA-266 2	155.000	20.000	...	...	...
N1- inlet	SA-266 2	155.000	20.000	...	...	...
N5	SA-266 2	45.900	15.000	...	...	...
Manhole	SA-266 2	62.900	15.000	...	...	...
Shell inlet	SA-266 2	46.000	10.000	...	...	...
Shell Outle	SA-266 2	46.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1- Tube inlet	750.000	0.0	500.00	0.00	Channel shell
N1- inlet	750.000	180.0	500.00	0.00	Channel shell
N5	...	0.0	500.00	0.00	Shell head
Manhole	...	0.0	500.00	0.00	Channel head
Shell inlet-N3	4713.000	0.0	300.00	0.00	Shell
Shell Outlet- N3	4713.000	180.0	300.00	0.00	Shell

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Bill of Material:

Step: 27 9:40am Apr 21,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 87.0mm THK X 2250.0mm ID X 0.0mm	SA-516 70
1	WELD NECK FLANGE: 340.0mm THK X 2680.0mm OD	SA-266 2
1	ELLIPTICAL HEAD: 2.0 X 1, 54.0mm THK X 2256.0mm ID X 50.0mm	SA-516 70
1	INSULATION: 612mm X 100mm THK	Hot
1	INSULATION: 1500mm X 100mm THK	Hot
4	BAFFLES: 2245.0mm DIA X 19.0mm THK	SA-516 70
12	TIE RODS: 16mm DIA X 5000mm	SA-193 B7M
1	TUBESHEET: 2680.0mm X 340.0mm THK	SA-266 2
4149	TUBES: 5000.0mm X 15.9mm DIA X 1.6mm THK	SA-213 TP316L
1	CLADDING: 430.000mm X 5.000mm THK	...
2	SADDLE: 204.0mm X 120 DEG	SA-516 70
1	INSULATION: 4900mm X 100mm THK	Hot
1	CLADDING: 4900.000mm X 3.000mm THK	...
1	INSULATION: 614mm X 100mm THK	Hot
1	CLADDING: 612.500mm X 3.000mm THK	...
1	CLASS 1500 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	GASKET: 2314.0mm OD X 2288.0mm ID	...
64	BODY FLANGE BOLTS: 57.1mm DIA	SA-193 B7
64	NUTS FOR BODY FLANGE BOLTS: 57.1mm DIA	...
2	CYLINDRICAL SEGMENT 155.0mm THK X 2807.8mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 155.0mm THK X 1453.0mm X 1500.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 46.0mm THK X 2728.6mm X 4900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 46.0mm THK X 1630.0mm X 4900.0mm	SA-516 70
1	NAMEPLATE	...



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MDMT Summary:

Step: 30 9:40am Apr 21,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Shell flng. [11]	B	-2	-6		0.923	46.000	1.00	Yes
Bolting [21]		-48						
Shell [8]	B	15	13		0.959	46.000	1.00	Yes
Shell head [10]	B	15	11		0.941	45.900	1.00	Yes
Shell head [7]	B	18	8		0.809	54.000	1.00	Yes
Left Saddle [24]	B	-31	-33		0.959	16.000	1.00	No
Left saddle [24]	B	-31	-33		0.959	16.000	1.00	No
Shell inlet-N3[1]	B	15	13		0.963	46.000	1.00	No
Shell Outlet- [1]	B	15	13		0.963	46.000	1.00	No
N5 [1]	B	15	6		0.851	45.900	1.00	No
Tubesheet: SS[13]	B	11	7		0.927	85.000	1.00	No
Warmest MDMT:		18	13					
Channel head [9]	B	25	25		1.000	73.950	1.00	Yes
Manhole [1]	B	35	32		0.939	47.000	1.00	Yes
Nozzle Flg [5]	A	-18	-36					
N1- Tube inlet[1]	B	23	5		0.676	29.159	1.00	Yes
N1- inlet [1]	B	23	5		0.676	29.159	1.00	Yes
Nozzle Flg [5]	A	-18	-36					
Warmest MDMT:		35	32					
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail			
Shell	13.0		-12.0		Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



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MDMT Summary:

Step: 30 9:40am Apr 21,2026

UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:40am Apr 21, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:40am Apr 21,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	4950	KPa.
Channel Side Design Pressure	17000	KPa.
Shell Side Design Temperature	275.0	°C
Channel Side Design Temperature	250.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 9:40am Apr 21,2026

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed [V]	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	27500	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration [Ss]	0.147	
Max. Eff. Ground Acceleration [S]	0.065	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio z/h	0.000	
Amplification Factor Ap	1.000	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

M.A.W.P. + Static Head to Element Bottom	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 9:40am Apr 21,2026

Use ASME VIII-1 Appendix 1-9 N
Perform Blast Load Analysis No
Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
Use Table G instead of exact equation for "A" Yes
Shell Head Joints are Tapered Yes
Compute "K" in corroded condition Yes
Use Code Case 2286 No
Use the MAWP to compute the MDMT Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes
Calculate B31.3 type stress for Nozzles with Loads Yes
Reduce the MDMT due to lower membrane stress Yes
Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5	
Element From Node	10	
Element To Node	20	
Element Type	Spherical	
Description	Channel head	
Distance "FROM" to "TO"	0	mm
Inside Diameter	2250	mm
Element Thickness	73.95	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	87	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	17000	KPa.
Design Temperature Internal Pressure	250	°C
Design External Pressure	101.3	KPa.
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	216	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-562.5	mm
Height/Length of Insulation	612.5	mm
Thickness of Insulation	100	mm
Density	0.00032	kg./cm ³
Element From Node	10	
Detail Type	Nozzle	



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Detail ID	Manhole
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	4187.1 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Channel shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	2250 mm
Element Thickness	155 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	155 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	17000 KPa.
Design Temperature Internal Pressure	250 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2250 mm
Liquid Density	0.0009355 kg./cm ³
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.00032 kg./cm ³
Element From Node	20
Detail Type	Nozzle
Detail ID	N1- Tube inlet
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	12 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1499.5 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-266 2
Element From Node	20
Detail Type	Nozzle
Detail ID	N1- inlet



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Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	12	in.
Nozzle Schedule	160	
Nozzle Class	1500	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1502.6	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	20	
Detail Type	Weight	
Detail ID	Pass partitin p	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	0	mm

Element#	3/5	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	Shell flng.	
Distance "FROM" to "TO"	370	mm
Flange Inside Diameter	2256	mm
Element Thickness	300	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	340	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4950	KPa.
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	132.5	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	199	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Lining	
Detail ID	claaing	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	430	mm
Thickness of Lining	5	mm
Density	0.008	kg./cm ³

Element#	4/5	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	4900	mm



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Inside Diameter	2256	mm
Element Thickness	46	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	46	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4950	KPa.
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	137	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	210	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1100	mm
Width of Saddle	204	mm
Height of Saddle at Bottom	1752	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	254	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.40000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left saddle	
Dist. from "FROM" Node / Offset dist	4300	mm
Width of Saddle	204	mm
Height of Saddle at Bottom	1752	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	254	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.40000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm



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Height/Length of Liquid	2256	mm
Liquid Density	0.0008926	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	4900	mm
Thickness of Insulation	100	mm
Density	0.00032	kg./cm ³

Element From Node	40	
Detail Type	Lining	
Detail ID	Weld overlay	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	4900	mm
Thickness of Lining	3	mm
Density	0.00785	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell inlet-N3	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	691.99	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell Outlet- N3	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	730.12	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element From Node	40	
Detail Type	Weight	
Detail ID	Fire proofing	
Dist. from "FROM" Node / Offset dist	1100	mm
Miscellaneous Weight	450	Kgf
Offset from Element Centerline	0	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	Fire proofing	
Dist. from "FROM" Node / Offset dist	4300	mm
Miscellaneous Weight	450	Kgf
Offset from Element Centerline	0	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	Skid bar	
Dist. from "FROM" Node / Offset dist	2700	mm
Miscellaneous Weight	450	Kgf
Offset from Element Centerline	0	mm



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Element From Node	40
Detail Type	Weight
Detail ID	Insulation clea
Dist. from "FROM" Node / Offset dist	2700 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	1000 mm

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	Shell head
Distance "FROM" to "TO"	50 mm
Inside Diameter	2256 mm
Element Thickness	45.9 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	54 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2256 mm
Liquid Density	0.0008926 kg./cm³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	614 mm
Thickness of Insulation	100 mm
Density	0.00032 kg./cm³

Element From Node	50
Detail Type	Lining
Detail ID	Weld overlay
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	612.5 mm
Thickness of Lining	3 mm
Density	0.00785 kg./cm³

Element From Node	50
Detail Type	Nozzle
Detail ID	N5
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	16 in.
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	270.42 Kg



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Grade of Attached Flange

None

Nozzle Matl

SA-266 2

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XY Coordinate Calculations: Step: 2 9:40am Apr 21,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Channel head		...	...	...	...
Channel shell		1500	...	1500	...
Shell flng.		2213	...	370	...
Shell		7113	...	4900	...
Shell head		7163	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: FLANGE

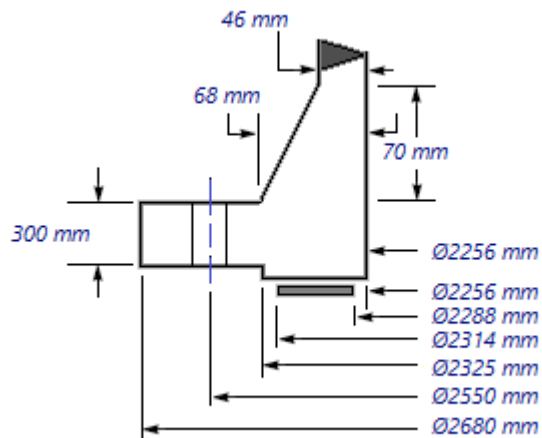
Flng: 3 9:40am Apr 21,2026

Flange Input Data Values

Description: FLANGE :

Shell flng.

Flange Type	Integral Weld Neck		
Design Pressure	P	4950.00	KPa.
Design Temperature		275	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	2256.000	mm
Flange Outside Diameter	A	2680.000	mm
Flange Thickness	t	300.0000	mm
Thickness of Hub at Small End	go	46.0000	mm
Thickness of Hub at Large End	gl	68.0000	mm
Length of Hub	h	70.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	132.50	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2550.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		64	
Flange Face Outside Diameter	Fod	2325.000	mm
Flange Face Inside Diameter	Fid	2256.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2314.000	mm
Gasket Inside Diameter	Gi	2288.000	mm
Gasket Factor	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Gasket Thickness	tg	3.0000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: FLANGE

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ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4950 \cdot (2256/2 + 3)) / (133 \cdot 1 - 0.6 \cdot 4950) + Ca$$

$$= 46.2235 \text{ mm} \quad \text{Warning: Less than specified Small End Hub thickness}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133 \cdot 1 \cdot 43) / (1131 + 0.6 \cdot 43)$$

$$= 4924.971 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 46) / (1128 + 0.6 \cdot 46)$$

$$= 5492.832 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B+2*ci	2262.000 mm
Corroded Large Hub,	g1Cor = g1-ci	65.000 mm
Corroded Small Hub,	g0Cor = go-ci	43.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	79.000 mm

Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm
Basic Gasket Width,	bo = N/2	6.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter,	G = Go - 2 * b	2301.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2301^2 \cdot 4950$$

$$= 2099335.750 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 2301 \cdot 4.25 \cdot 4950$$

$$= 199277.812 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 2262^2 \cdot 4950$$

$$= 2028508.875 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 2099335 - 2028508$$

$$= 70826.914 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(2099335 + 199278 + 0, 0)$$

$$= 2298613.750 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 69.6 \cdot 6.42 \cdot \pi \cdot 2301 + 0 \cdot 0$$

$$= 329811.062 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((2298613 + 0 + 4 \cdot 0 / 2301) / 172, 329811 / 172)$$

$$= 1310.612 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:



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Flg Calc [Int P]: FLANGE

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$$\begin{aligned}
 &= 2a + 6t/(m + 0.5) \\
 &= 2 \cdot 57.1 + 6 \cdot 300/(4.25 + 0.5) \\
 &= 493.247 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 2550 \cdot \sin(\pi / 64) \\
 &= 125.123 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{(Bs/(2a + t))}, 1) \\
 &= \max(\sqrt{(125/(2 \cdot 57.1 + 300))}, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1310.612	1413.365	
Radial Distance between Hub and Bolts:	69.850	79.000	
Radial Distance between Bolts and Edge:	57.150	65.000	
Circ. Spacing between the Bolts:	120.650	125.123	493.247

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= Ab \cdot Sa / (y \cdot \pi(Go + Gi)) \\
 &= 1413 \cdot 172 / (69.6 \cdot \pi(2314 + 2288)) \\
 &= 24.145 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000}]
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= Sa(Am + Ab) / 2 \\
 &= 172(1311 + 1413) / 2 \\
 &= 2388720.25 \text{ Kg}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= Wo - H \\
 &= 2298613 - 2099335 \\
 &= 199277.81 \text{ Kg}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (2550 - 2301) / 2 \\
 &= 124.4245 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5((C - B) / 2 + hg) \\
 &= 0.5((2550 - 2262) / 2 + 124) \\
 &= 134.2122 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g1) / 2 \\
 &= (2550 - 2262 - 65) / 2 \\
 &= 111.4999 \text{ mm}
 \end{aligned}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	2028509.	111.4999	1.0000	226183.



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Flg Calc [Int P]: FLANGE

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Face Pressure, Mt	70827.	134.2122	1.0000	9506.
Gasket Load, Mg	199278.	124.4245	1.0000	24796.
Gasket Seating, Matm	2388720.	124.4245	1.0000	297221.

Total Moment for Operation, Mo 260485. Kg-m.
Total Moment for Gasket seating, Mg 297221. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 311.875 mm
Hub Ratio, h/h0 = HL / H0 0.224
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.512

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.896
Factor V per 4.16.5 0.399
Factor f per 4.16.5 1.439

Stress factor from Table 4.16.4 [K]:

= A / B
= 2680/2262
= 1.185

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169(1.18² • log10(1.18)) / (1.18² - 1)) / (1.18 - 1)
= 11.539

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.18²(1 + 8.55246 • log10(1.18)) - 1) / ((1.0472 + 1.9448 • 1.18²) • (1.18 - 1))
= 1.845

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.18²(1 + 8.55246 • log10(1.18)) - 1) / (1.36136(1.18² - 1) • (1.18 - 1))
= 12.680

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.18² + 1) / (1.18² - 1)
= 5.954

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 12.7 • 43² • 312 / 0.4
= 18310588.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.9/312
= 0.003 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (300 • 0.0029 + 1) / 1.85 + 300³/18310588
= 2.484

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.44•260485/2262)/(2.48•65²)
= 154.86 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)



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Flg Calc [Int P]: FLANGE

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$$= (1.44 \cdot 297221 / 2262) / (2.48 \cdot 65^2)$$

$$= 176.70 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.15 \cdot 260485 / 2262) / (2.48 \cdot 300^2)$$

$$= 10.86 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.15 \cdot 297221 / 2262) / (2.48 \cdot 300^2)$$

$$= 12.39 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (11.5 \cdot 260485 / (300^2 \cdot 2262)) - 5.95 \cdot 10.9$$

$$= 80.15 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (11.5 \cdot 297221 / (300^2 \cdot 2262)) - 5.95 \cdot 12.4$$

$$= 91.45 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (155 + \max(10.9, 80.1)) / 2$$

$$= 117.51 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (177 + \max(12.4, 91.4)) / 2$$

$$= 134.08 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 2298613 / 1413$$

$$= 159.49 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (329811 / 1413)$$

$$= 22.88 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	154.86	198.75	176.70	207.00
Radial Flange	10.86	132.50	12.39	138.00
Tangential Flange	80.15	132.50	91.45	138.00
Maximum Average	117.51	132.50	134.08	138.00
Bolting	159.49	172.00	22.88	172.00

Minimum Required Flange Thickness 294.564 mm

Estimated M.A.W.P. (Operating) 4925 KPa.

Estimated Finished Weight of Flange at given Thk. 4046.6 kg.

Estimated Unfinished Weight of Forging at given Thk 4713.7 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

Operating stress	Gasket Seating stress
------------------	-----------------------



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Flg Calc [Int P]: FLANGE

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		Actual	Allowed	ratio	Actual	Allowed	ratio
Longitudinal Hub	Sh	454.86	454.86	1.000	454.86	499.99	0.910
Tangential Flange	St	237.76	341.14	0.697	237.76	374.99	0.634
Tangential Flange	Sto	96.90	227.43	0.426	96.90	250.00	0.388
Combined	Sh/f + St	557.03	682.29	0.816	557.03	749.99	0.743
Combined	Sr + St	269.96	682.29	0.396	269.96	749.99	0.360
Combined	Sh + Sto	551.76	682.29	0.809	551.76	749.99	0.736
Radial Flange	Sr	32.21	454.86	0.071	32.21	499.99	0.064
Maximum Stress Ratio				1.000			0.910

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.59/1.12)

= 1.4247

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)

= max(0.25; 0.31/1.01) if h/ho < 0.35, else max (0.25; 0.27/0.79)

= 0.3035

In this case, the hub ratio h/ho, 0.2244 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



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Flg Calc [Int P]: FLANGE

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Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	430.9 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	340.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	22.0838 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	199.0 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.463 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2314^2 - 2288^2)$$

$$= 939.7 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 208 \cdot 940 / (64 \cdot 22.1)$$

$$= 138.3 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(138; 508)$$

$$= 138.3 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(138; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 431)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (940 / (64 \cdot 22.1))$$

$$:: 290.0 \geq 93.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 940 + \pi/4 \cdot 4950 \cdot 2288^2) / (0.7 \cdot 22.1 \cdot 64)$$

$$:: 290.0 \geq 297.9 \text{ N./mm}^2 \text{ Check Fails}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 290 \leq 340 (940 / (64 \cdot 22.1))$$

$$:: 290.0 \leq 226.1 \text{ N./mm}^2 \text{ Check Fails}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetaamax)$$

$$:: 290 \leq 431 (1/0.46)$$

$$:: 290.0 \leq 930.9 \text{ N./mm}^2 \text{ Check Passes}$$



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Flg Calc [Int P]: FLANGE

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Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.2 \cdot 22.1 \cdot 57.1$$

$$= 746.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 87 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 297221/1 \cdot 9807 \cdot 0.4 / (2.4 \cdot 201334 \cdot 43^2 \cdot 312 \cdot 0.3)$$

$$= 0.727 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 260485/1 \cdot 9807 \cdot 0.4 / (2.4 \cdot 185553 \cdot 43^2 \cdot 312 \cdot 0.3)$$

$$= 0.691 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 297221/1 \cdot 9807 \cdot 0.4 / (2.48 \cdot 201334 \cdot 43^2 \cdot 312 \cdot 0.3)$$

$$= 0.701 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 260485/1 \cdot 9807 \cdot 0.4 / (2.48 \cdot 185553 \cdot 43^2 \cdot 312 \cdot 0.3)$$

$$= 0.667 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.92, Temperature Reduction per Fig. UCS 66.1 = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	151.80	207.00	170.98	207.00
Radial Flange	11.77	138.00	13.26	138.00
Tangential Flange	75.82	138.00	85.40	138.00
Maximum Average	113.81	138.00	128.19	138.00
Bolting	159.49	172.00	22.88	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 285.420 mm

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Internal Pressure Calculations: Step: 4 9:40am Apr 21,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Channel head		17000	87	3	2250	138	1
Channel shell		17021	155	3	2250	138	2
Shell flng.		4950	340	3	2256	132.5	3
Shell		4969.8	46	3	2256	137	4
Shell head		4969.8	54	3	2256	137	5

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Channel head		17000	17143	17906	73.95	73.3503	1
Channel shell		17000	17183	17560	155	153.257	2
Shell flng.		4950 >>>	4925	5338	300	294.564	3
Shell		4950	5072	5493	46	44.9439	4
Shell head		4950	5175	5592	45.9	44.0352	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 4925 KPa., < design pressure: 4950.000 KPa.
 Shell Side MAPnc = 5338 KPa.
 Channel Side MAWP = 17143 KPa.
 Channel Side MAPnc = 17560 KPa.

Elements Suitable for Design Internal Pressure.

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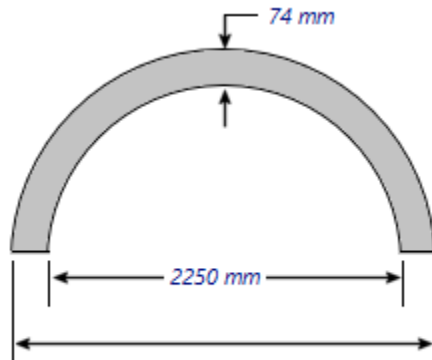
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Element: Channel head

Step: 5 9:40am Apr 21,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel head

Spherical Head Design Information:			
Design Pressure P	17000.002 KPa.	Int. Design Temperature	250 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	0.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2250.000 mm	Actual Stress	136.8 N./mm ²
Spec. Min. Thk t _e , or s _a	73.9500 mm	Spec. Nominal Thk.	87.000 mm
Surface Area	92296.5 cm ²	SG of Contents	0.000
Element Volume	2982593.2 cm ³	Weight of Contents	0.0 Kgf
Empty Weight	5787.4 Kgf	Operating Weight	5787.4 Kgf
		Comp. All. Ope. Stress	120.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.125 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.062 mm

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\
 &= (17000 \cdot 1128) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 17000) \\
 &= 70.3503 + 3.0000 = 73.3503 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)} \\
 &= (2 \cdot 138 \cdot 1 \cdot 71) / (1128 + 0.2 \cdot 71) \\
 &= 17143 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)} \\
 &= (2 \cdot 138 \cdot 1 \cdot 74) / (1125 + 0.2 \cdot 74)
 \end{aligned}$$



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Element: Channel head

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= 17906 KPa.

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (17000 \cdot (1128 + 0.2 \cdot 71)) / (2 \cdot 1 \cdot 71)$$

$$= 136.845 \text{ N./mm}^2$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.584 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 74$, $t_r = 71$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 25 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	17143.129 KPa.
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	73.950 mm	Basic MDMT	25 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	25 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.125 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.062 mm

EXTERNAL PRESSURE RESULTS for: Channel head

Spherical Head From 10 to 20 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
70.950	2397.90	33.80	0.0073971	111.53

$$MAEP = B / ((Do/t)/2) = 112 / (33.8/2) = 6600 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
3.469	2397.90	691.20	0.0003617	35.01

$$MAEP = B / ((Do/t)/2) = 35 / (691/2) = 101 \text{ KPa.}$$



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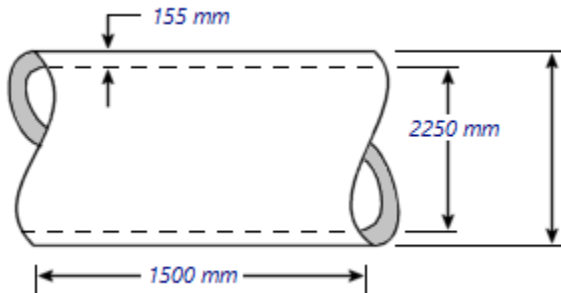
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Element: Channel shell

Step: 6 9:40am Apr 21,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	17020.693 KPa.	Int. Design Temperature	250 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2250.000 mm	Actual Stress	136.5 N./mm ²
Spec. Min. Thk t,e, or sa	155.0000 mm	Spec. Nominal Thk.	155.000 mm
Surface Area	120637.2 cm ²	SG of Contents	0.937
Element Volume	5965186.5 cm ³	Weight of Contents	5586.0 Kg
Empty Weight	13615.0 Kg	Operating Weight	19201.0 Kg
		Comp. All. Ope. Stress	131.3 N./mm ²
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (17021 \cdot 1128) / (138 \cdot 1 - 0.6 \cdot 17021) \\
 &= 150.2565 + 3.0000 = 153.2565 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 20.693 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 152) / (1128 + 0.6 \cdot 152) \\
 &= 17203 - 20.7 = 17183 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 155) / (1125 + 0.6 \cdot 155) \\
 &= 17560 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (17021 \cdot (1128 + 0.6 \cdot 152)) / (1 \cdot 152) \\
 &= 136.532 \text{ N./mm}^2
 \end{aligned}$$



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Element: Channel shell

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% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 6.445 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	17143.129 KPa.
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	155.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Channel shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
152.000	2560.00	1875.00	16.84	0.7324	0.0301954	120.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*120)/(3*16.8) = 9515 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.220	2560.00	1875.00	354.57	0.7324	0.0002783	26.94

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.9)/(3*355) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
152.000	2560.00	164210.64	16.84	50.0000	0.0039955	103.92

$$MAEP = (4*B)/(3*(Do/t)) = (4*104)/(3*16.8) = 8227 \text{ KPa.}$$

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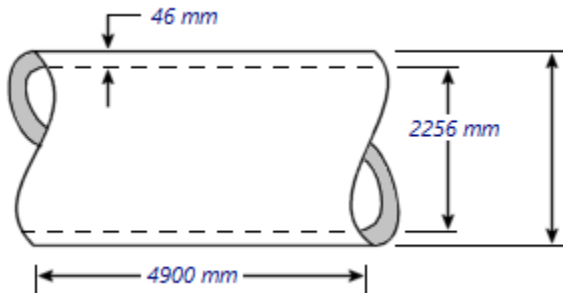
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Element: Shell

Step: 7 9:40am Apr 21,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	4969.796 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	4900.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2256.000 mm	Actual Stress	133.7 N./mm ²
Spec. Min. Thk t,e, or sa	46.0000 mm	Spec. Nominal Thk.	46.000 mm
Surface Area	361446.5 cm ²	SG of Contents	0.894
Element Volume	14880526.0 cm ³	Weight of Contents	13295.1 Kg
Empty Weight	12633.9 Kg	Operating Weight	25929.0 Kg
		Comp. All. Ope. Stress	106.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4970 \cdot 1131) / (137 \cdot 1 - 0.6 \cdot 4970) \\
 &= 41.9439 + 3.0000 = 44.9439 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 19.796 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137 \cdot 1 \cdot 43) / (1131 + 0.6 \cdot 43) \\
 &= 5092 - 19.8 = 5072 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 46) / (1128 + 0.6 \cdot 46) \\
 &= 5493 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4970 \cdot (1131 + 0.6 \cdot 43)) / (1 \cdot 43) \\
 &= 133.707 \text{ N./mm}^2
 \end{aligned}$$

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Element: Shell

Step: 7 9:40am Apr 21, 2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.998 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 46$, $t_r = 41.2$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $tr * E^*/(tg - c) = 0.96$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C

Min Metal Temp. at Required thickness (UCS 66.1)	13 °C
--	-------

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4924.971 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	46.000 mm	Basic MDMT	15 °C
Stress/Thk. Ratio	0.959	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	13 °C
		Joint Efficiency E*	100 %

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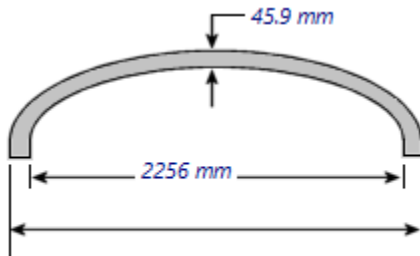
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Element: Shell head

Step: 8 9:40am Apr 21,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell head

Elliptical Head Design Information:			
Design Pressure P	4969.797 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2256.000 mm	Actual Stress	131.1 N./mm ²
Spec. Min. Thk t,e, or sa	45.9000 mm	Spec. Nominal Thk.	54.000 mm
Surface Area	64291.8 cm ²	SG of Contents	0.894
Element Volume	1703160.2 cm ³	Weight of Contents	1521.7 Kg
Empty Weight	2824.8 Kg	Operating Weight	4346.5 Kg
		Comp. All. Ope. Stress	106.7 N./mm ²
		Aspect Ratio	2.00
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.200 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.100 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4970 \cdot 2262 \cdot 1) / (2 \cdot 137 \cdot 1 - 0.2 \cdot 4970)$$

$$= 41.0352 + 3.0000 = 44.0352 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 19.796 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137 \cdot 1 \cdot 42.9) / (1 \cdot 2262 + 0.2 \cdot 42.9)$$

$$= 5195 - 19.8 = 5175 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 45.9) / (1 \cdot 2256 + 0.2 \cdot 45.9)$$

$$= 5592 \text{ KPa.}$$



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Element: Shell head

Step: 8 9:40am Apr 21,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4970 \cdot (1 \cdot 2262 + 0.2 \cdot 42.9)) / (2 \cdot 1 \cdot 42.9)$$

$$= 131.065 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4970 \cdot 1131) / (137 \cdot 1 - 0.6 \cdot 4970) + 3$$

$$= 44.944 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 19.796 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 51) / (1131 + 0.6 \cdot 51)$$

$$= 6015 - 19.8 = 5995 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (2262 / (2 \cdot 567)))^2 / 6$$

$$= 0.996477$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.866 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 45.9, tr = 40.4, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.94$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C

Min Metal Temp. at Required thickness (UCS 66.1) 11 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 54, tr = 41.2, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.81$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 18 °C

Min Metal Temp. at Required thickness (UCS 66.1) 8 °C

[Note:](#)

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4924.971 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	54.000 mm	Basic MDMT	18 °C
Stress/Thk. Ratio	0.809	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	8 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.200 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.100 mm



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Shop Pressure Test Results: Step: 9 9:40am Apr 21,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6449.152 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6481.927 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6917.397 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5456.975 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	4924.971 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	22286.064 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	22100.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	22806.195 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	18857.441 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	24310.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	24310.000 KPa.
App 27-4	= M.A.W.P.	17143.129 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4950 • 1.01

= 6481.927 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 17000 • 1

= 22100.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel head	170.5	234.0	0.729	22122.05
Channel shell	173.8	234.0	0.743	22122.05
Shell	163.4	234.0	0.698	6504.04
Shell head	160.5	234.0	0.686	6504.04

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	138.00	136.00	1.015
N1- Tube inlet	Nozzle	138.00	136.00	1.015
N1- inlet	Nozzle	138.00	136.00	1.015
Shell inlet-N3	Nozzle	138.00	132.50	1.041
Shell Outlet- N3	Nozzle	138.00	132.50	1.041
N5	Nozzle	138.00	132.50	1.041

Minimum

1.015

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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Channel head	138.00	138.00	1.000
Channel shell	138.00	138.00	1.000
Shell flng.	138.00	132.50	1.041
Shell	138.00	137.00	1.007
Shell head	138.00	137.00	1.007

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	103.00	1.116
Tubesheet Material	138.00	136.00	1.015

Minimum 1.015

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Manhole	156.75	225.00	0.697
N1- Tube inlet	98.65	225.00	0.438
N1- inlet	98.65	225.00	0.438
Shell inlet-N3	68.20	225.00	0.303
Shell Outlet- N3	96.53	225.00	0.429
N5	30.04	225.00	0.134

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 10 9:40am Apr 21,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6449.152 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6481.927 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6917.397 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5456.975 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	4924.971 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	22286.064 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	22100.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	22806.195 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	18857.441 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	24310.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	24310.000 KPa.
App 27-4	= M.A.W.P.	17143.129 KPa.

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel head	179.6	234.0	0.767	22308.12
Channel shell	178.9	234.0	0.765	22308.12
Shell	174.1	234.0	0.744	6471.27
Shell head	170.7	234.0	0.729	6471.27

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	138.00	136.00	1.015
N1- Tube inlet	Nozzle	138.00	136.00	1.015
N1- inlet	Nozzle	138.00	136.00	1.015
Shell inlet-N3	Nozzle	138.00	132.50	1.041
Shell Outlet- N3	Nozzle	138.00	132.50	1.041
N5	Nozzle	138.00	132.50	1.041

Minimum 1.015

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Channel head	138.00	138.00	1.000
Channel shell	138.00	138.00	1.000
Shell flng.	138.00	132.50	1.041
Shell	138.00	137.00	1.007
Shell head	138.00	137.00	1.007

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):



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Field Pressure Test Results: Step: 10 9:40am Apr 21,2026

Description	Ambient	Operating	Ratio
Tube Material	115.00	103.00	1.116
Tubesheet Material	138.00	136.00	1.015

Minimum 1.015

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Manhole	158.06	225.00	0.703
N1- Tube inlet	99.48	225.00	0.442
N1- inlet	99.48	225.00	0.442
Shell inlet-N3	67.86	225.00	0.302
Shell Outlet- N3	96.04	225.00	0.427
N5	29.89	225.00	0.133

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		6449.15 KPa.
Shell Thickness	ts	46.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2256.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Test Pressure		22286.06 KPa.
Channel Thickness	tc	155.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2250.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4149
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	15.8800 mm
Total Straight Tube Length	Lt	5000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		4660.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP316L
Tube Material UNS Number		S31603
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	170.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	22.2300 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	892.044 mm
Largest Center-to-Center Tube Distance	U1	50.0000 mm
Length of Expanded Portion of Tube	ltx	100.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	24327.8 cm ²
Perimeter of Tube Layout	Cp	7620.6 mm



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ASME TS Calc: Pressure Test

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Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	340.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2680.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	2325.00 mm
Step 1 Depth on the Shellside	hs1	9.00 mm
Step 2 Diameter on the Shellside	ds2	2256.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2680.00 mm

Note: Tubesheet diameter is now: 2680.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	3375.0 cm ²
---------------------------	----	------------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2680.000 mm
Flange Inside Diameter	B	2256.000 mm
Flange Face Outside Diameter	Fod	2325.000 mm
Flange Face Inside Diameter	Fid	2256.000 mm
Gasket Outside Diameter	Go	2314.000 mm
Gasket Inside Diameter	Gi	2288.000 mm
Small end Hub thk.	g0	46.0000 mm
Large end Hub thk.	g1	68.0000 mm
Gasket Factor,	m	4.25
Gasket Design Seating Stress	y	69.64 N./mm ²
Flange Facing Sketch	Code	Sketch 1a
Gasket Thickness	tg	3.0000 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2550.000 mm
Nominal Bolt Diameter	dB	57.1500 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	64
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Tubesheet Integral with		Channel
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	324.0000 mm



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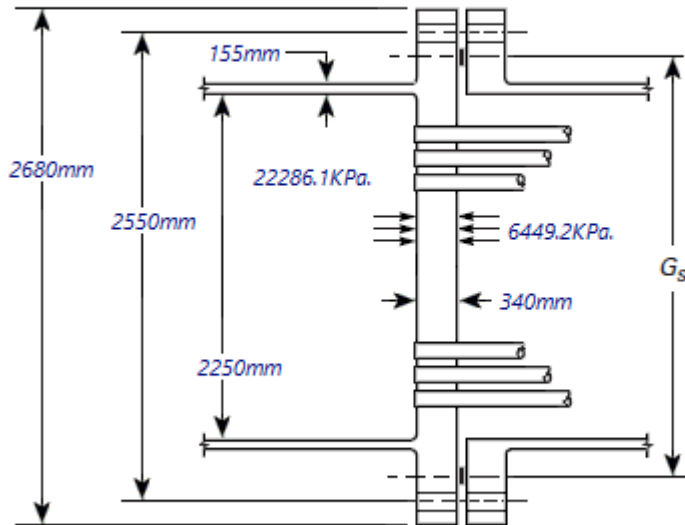
ASME TS Calc: Pressure Test

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Is Bolt Load Transferred to the Tubesheet

Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesheet Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (22286 \cdot (15.9/2 - 0)) / (115 \cdot 1 + 0.4 \cdot 22286)$
 $= 1.4281 + 0.0000 = 1.4281 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-4 at 21.11 °C



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Elastic Modulus for Material 193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	15.88	5552.04	9.62	50.0000	0.0118757	97.63

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 13849$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.9231	15.88	5552.04	17.20	50.0000	0.0037171	83.22

EMAWP = $(4 * B) / (3 * (D/T)) = (4 * 83.2) / (3 * 17.2) = 6450$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4281 mm
Allowable Internal Pressure at Corroded thickness	26062.57 KPa.
Required Internal Design Pressure	22286.06 KPa.
Allowable External Pressure at Corroded thickness	13848.94 KPa.
Required External Design Pressure	6449.15 KPa.
Required Thickness due to Shell Side pressure	0.9231 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	2301.156 mm

Warning: Available bolt area, 1413 cm² for the flange is < required 1707 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * ro + dt$
 $= 2 * 892 + 15.9$
 $= 1799.968$ mm

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (22.2 - 15.9) / 22.2$
 $= 0.286$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Gs / Do$ (Configurations d, e, f)
 $= 2301 / 1800$
 $= 1.2784$

Ratio [pc]:

$= Dc / Do$ (Configurations a, e, f)
 $= 2250 / 1800$



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$$= 1.2500$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 100/340$$

$$= 0.294 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t(E_t/E) \cdot (StT/S) \cdot (p); dt - 2t)$$

$$= \max(15.9 - 2 \cdot 1.65 (195117184/202701232) \cdot$$

$$(115/138) \cdot (0.29); 15.9 - 2 \cdot 1.65)$$

$$= 15.1014 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 22.2 / \sqrt{ (1 - 4 \cdot \min(3375 ; 4 \cdot 1800 \cdot 22.2) / (\pi \cdot 1800^2)) }$$

$$= 22.9639 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (23 - 15.1) / 23$$

$$= 0.3424$$

$$h/p = 15.294647 ; \mu^* = 0.342385 ; E^* = 68326296 \text{ KPa.}$$

$$E^*/E = 0.337079 ; \nu^* = 0.329852$$

Note: As h/p (15.295) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

βs, ks, λs, δs and ωs are all = 0

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2250 + 155) 155 }$$

$$= 0.0030 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.003 \cdot 201322320 \cdot 155^3 / (6(1 - 0.3^2))$$

$$= 408814813184.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta +$$

$$h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2250 + 155 / \cos(0)) \cdot 408.81E+09 / 340^3 \cdot (1 +$$

$$340 \cdot 0.003 \cdot (1 + 1) / 2 + (340 \cdot 0.003)^2 / 2 \cdot 1)$$

$$= 378933792.0 \text{ KPa.}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2250^2 / (4 \cdot 201322320 \cdot 155) \cdot (1 - 2250 \cdot 0.3 / (2(2250 + 155)))$$

$$= 0.34865E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$= p_c \cdot kc [\delta C(CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) -$$



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$$\begin{aligned} & \zeta_c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta_C / 2) \\ &= 1.25 \cdot 408.81 \text{E}+09 [0.034864757 (1 + 0.003 \cdot 1 \cdot 340) - \\ & \quad 0.000 (1/340 + 1 \cdot 0.003/2)] \\ &= 106753.4922 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 2680/1800 \\ &= 1.4889 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.33) / 68326296 \cdot (378933792 + 202701232 \cdot \ln(1.49)) \\ &= 4.5080 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t] \\ &= 1800^2 / 16 \cdot [(1.28 - 1) \cdot (1.28^2 + 1) \cdot 6449 - \\ & \quad (1.25 - 1) \cdot (1.25^2 + 1) \cdot 22286] \\ &= -1933384704.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= M_{ts} + (\omega_c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o) \\ &= -1933384704 + (2.38 \text{E}+09) + 2994685 \cdot (2550 - 2301) / (2 \cdot \pi \cdot 1800) \\ &= 1091887232.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= (M^* - D_o^2 / 32 \cdot F (P_s - P_t)) / (1 + F) \\ &= ((1091887232) - 1800^2 / 32 \cdot 4.51 (6449 - 22286)) / (1 + 4.51) \\ &= 1510556928.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t) \\ &= 1510556928 + 1800^2 / 64 \cdot (3 + 0.33) \cdot (6449 - 22286) \\ &= -1159036288.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(|1510556928|; |-1159036288|) \\ &= 1510556928.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 1510556928 / (0.34 (340 - 5)^2) \\ &= 235.8901 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot \beta_T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 250 \\ &= 474.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 1514926336 / (0.36 (237 - 5)^2) \\ &= 474.7302 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:



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$$\begin{aligned}
 &= h + \text{Cats} + \text{Catc} \\
 &= 237 + 0 + 0 \\
 &= 236.5232 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(\text{HreqB}; \text{HreqS}) \\
 &= \max(237 ; 93.2) \\
 &= 236.5232 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= | \text{Ps} - \text{Pt} | \\
 &= | 6449 - 22286 | \\
 &= 15836.913 \text{ KPa.}
 \end{aligned}$$

Shear Stress check [τ limit]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / \text{Do} \cdot 0.8 \cdot \beta T \cdot S_y \\
 &= 4 \cdot 0.29 \cdot 340 / 1800 \cdot 0.8 \cdot 0.95 \cdot 250 \\
 &= 41.007 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc} \\
 &= 1 / (4 \cdot 0.29 \cdot 190) \cdot (4 \cdot 24328 / 7621) \cdot | 6449 - 22286 | + 0 \\
 &= 93.1599 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 155.000 mm

Min. Channel len. adjacent to the tubesheet is: 1062.991mm

Tube Metal Area [At]:

$$\begin{aligned}
 &= \pi (\text{do} - t) t \\
 &= \pi (15.9 - 1.65) 1.65 \\
 &= 0.738 \text{ cm}^2
 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned}
 &= \pi \cdot t (\text{do} - t) S \\
 &= \pi \cdot 1.65 (15.9 - 1.65) 161 \\
 &= 1214.716 \text{ Kg}
 \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (\text{do} + 0.67 \cdot a_g) S_w \text{ (but not } > \text{ Ft)} \\
 &= 0.85 \cdot \pi \cdot 3 (15.9 + 0.67 \cdot 3) 161 \\
 &= 1214.7158 \text{ Kg}
 \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 161 / 161 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned}
 &= 0.55 \cdot \pi \cdot a_f (\text{do} + 0.67 \cdot a_f) S_w \text{ (but not } > \text{ Ft)} \\
 &= 0.55 \cdot \pi \cdot 3 (15.9 + 0.67 \cdot 3) 161 \\
 &= 1214.7158 \text{ Kg}
 \end{aligned}$$



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Minimum Length of Weld [ar]:

$$= 2(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do})$$

$$= 2(\sqrt{((0.75 \cdot 15.9)^2 + 1.07 \cdot 1.65 \cdot (15.9 - 1.65) \cdot 1 \cdot 1) - 0.75 \cdot 15.9})$$

$$= 2.023 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0117 mm

Minimum Required Groove Weld Leg [agr]: 1.0117 mm

Tube-Tubesheet Jt allowable, 1215 is >= tube strength 1215 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	235.9 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	52.1 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	236.523	340.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.012	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.012	3.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	264.908 112.711	313	475 73 190	Ps+Pt D1	Ok
D2uc	141.288 ...	132	475 21 190	Ps+Pt D2	Ok
D3uc	236.523 ...	236	475 52 190	Ps+Pt D3	Ok
Max:	264.9076 112.711 mm	0.659	0.386 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass/Fail
D1uc	...	...	...	...	370	Ok
D2uc	...	...	...	...	370	Ok



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ASME TS Calc: Pressure Test

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D3uc	...	...	...	...	...	370	Ok
------	-----	-----	-----	-----	-----	-----	----

Max	...	...	...	...
-----	-----	-----	-----	-----

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	264.908	340.000	Ok
Tubesheet thickness flanged extension :	112.711	324.000	Ok
Tube thickness :	1.428	1.650	Ok
Tube-tubesheet fillet weld leg :	1.012	3.000	Ok
Tube-tubesheet groove weld leg :	1.012	3.000	Ok

Given Shell Thickness: 46.0000 mm

Given Channel Thickness: 155.0000 mm

Min Channel length of thk, (155.000) adj. to tubesheet: 1062.991 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculations: Step: 11 9:40am Apr 21,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	2397.9	70.95	0.0073971	111.531
20	30	1875	2560	152	0.030195	120.195
30	40	No Calc	...	297	No Calc	No Calc
40	50	5138	2348	43	0.0014492	95.8892
50	60	No Calc	2347.8	42.9	0.0025378	108.78

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	73.95	6.46917	101.3	6600
20	30	155	10.2199	101.3	9515
30	40	300	194.513	...	No Calc
40	50	46	No Calc	...	2341
50	60	45.9	No Calc	...	2208

Minimum

2208

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	1875	164211	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	5138	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 12

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	5787.45	2982593	5602.05	3006517	578.745
20	30	13615	5965186	13368.1	5997043	1361.5
30	40	4556.23	1646432	4488.57	1647922	455.623
40	50	12633.9	14880526	11825.4	14984868	1263.39
50	60	2824.8	1703160	2667.87	1716251	282.48
Total		39417	27177900.00	37952	27352602.00	3941



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Ins1	347.717	-256.25	...	...	Ins: 10
10	Noz1	4605.79	-1125	...	...	Manhole
20	Liqd	5585.97	750	...	...	Liquid: 20
20	Ins1	401.588	750	...	...	Ins: 20
20	Noz1	1649.43	750	1253.6	...	N1- Tube inlet
20	Noz1	1652.83	750	-1253.6	...	N1- inlet
20	Wght	250	1000	...	...	Pass partitin p
30	Lini	121.775	215	...	...	cladding
40	Sad1	936.02	1100	1409	...	Left Saddle
40	Sad1	936.02	4300	1409	...	Left saddle
40	Liqd	13295.1	2450	...	...	Liquid: 40
40	Ins1	1207.3	2450	...	...	Ins: 40
40	Lini	817.73	2450	...	...	Weld overlay
40	Noz1	761.189	2500	1509	...	Shell inlet-N3
40	Noz1	803.131	2500	-1509	...	Shell Outlet- N3
40	Wght	450	1100	...	...	Fire proofing
40	Wght	450	4300	...	...	Fire proofing
40	Wght	450	2700	...	...	Skid bar
40	Wght	200	2700	1000	...	Insulation clea
50	Liqd	1521.75	238	...	...	Liquid: 50
50	Ins1	270.516	307	...	...	Ins: 50
50	Lini	151.583	306.25	...	...	Weld overlay
50	Noz1	297.465	614	...	...	N5
20	FTsh	13969.7	1670	...	...	Tubesheet
20	Tube	16377	4000	...	...	Tubes
30	Baf1	536.085	1250	...	...	Baffles
30	Baf1	536.085	2500	...	...	Baffles
30	Baf1	536.085	3750	...	...	Baffles
30	Baf1	536.722	5000	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	1872.0
Liquid	20402.8
Insulation	2227.1
Cladding & Overlay	1091.1
Nozzles	9769.8
Weights	1800.0



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Element and Detail Weights: Step: 12 9:40am Apr 21,2026

Exchanger Components	32491.7
Liquid in Tubes	3319.9

Sum of the Detail Weights 72974.4 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	43359.2	43359.2	43359.2	43359.2	43359.2	43359.2
Saddles	1872.0	1872.0	1872.0	1872.0	1872.0	1872.0
Cladding	1091.1	1091.1	1091.1	1091.1	1091.1	1091.1
Nozzles	9769.8	9769.8	9769.8	9769.8	9769.8	9769.8
Wld Weights	450.0	450.0	450.0	450.0	450.0	450.0
Exchanger	32491.7	32491.7	32491.7	32491.7	32491.7	32491.7
Insulation	...	...	...	2227.1	2227.1	2227.1
Empty Weights	...	...	...	1350.0	1350.0	...
Test Weights	...	1350.0	...	...	...	...
Ope Weights	...	...	...	...	...	1350.0
Ope. Liquid	...	...	...	...	...	20402.8
Tube Ope Lqd	...	...	...	...	...	3319.9
Test Liquid	...	27022.8	...	...	...	...
Tube Tst Lqd	...	3319.9	...	...	...	...
Totals	89033.8	120726.5	89033.8	92610.9	92610.9	116333.6

Field Installation Options:

Insulation installed before lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	89033.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	120726.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	89033.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	92610.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	92610.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	116333.6 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	114721.7 kg.
Field Test Wt.	- Empty weight + test medium (full)	122953.6 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	3321936.8 cm ³
Weight of Ope Liq in tubes :	3319.9 kg.
Weight of Water in tubes :	3319.9 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	92296.5	79521.6
20	30	120637	106029
30	40	50323.8	29058.5
40	50	361447	347284
50	60	64291.8	58713.5

Total	688995.9	620606.6 cm ²
Total	741.6	668.0 ft ²



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Element and Detail Weights:

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Nozzle Flange MAWP:

Step: 13 9:40am Apr 21,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Manhole	20980.2	25510.0	250	1500	GR 1.1	...	...	...	...
N1- inlet	20980.2	25510.0	250	1500	GR 1.1	...	...	...	...

Shellside Flange Rating**Channelside Flange Rating**

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 20980.186 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 25509.998 KPa.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange.

See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation:

Step: 14 9:40am Apr 21,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	27500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.18 \cdot 0.91)/(1 + 1.7 \cdot 3.4 \cdot 0.18)))$$

$$= \min(0.85, 0.88)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.538
Structure Height to Diameter ratio	3.261

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

= Centroid Height + Vessel Base Elevation
 = 1752 + 27500 = 29252 mm
 = 96 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (96 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (96/900)^{2/9.5}$$

$$= 1.25$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 14 9:40am Apr 21,2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.25 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.54 • 28.5

= 209 lbs. [94.6] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Channel head	29252.0	0.000	0.000	0.000	1.255	1.000	78.120
Channel shell	29252.0	0.000	0.000	0.000	1.255	1.000	78.120
Shell flng.	29252.0	0.000	0.000	0.000	1.255	1.000	78.120
Shell	29252.0	0.000	0.000	0.000	1.255	1.000	78.120
Shell head	29252.0	0.000	0.000	0.000	1.255	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Pass partitin	0.00	29252.00	78.12	0.00
Fire proofing	0.00	29252.00	78.12	0.00
Fire proofing	0.00	29252.00	78.12	0.00
Skid bar	0.00	29252.00	78.12	0.00
Insulation cl	0.00	30252.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	29252	3117.48	26504.5	78.12	104.093
20	30	29252	3312	49680	78.12	195.111
30	40	29252	2707.2	10016.6	78.12	39.3389
40	50	29252	3057.6	149822	78.12	588.405
50	60	29252	3057.36	15734	78.12	61.7931

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 15 9:40am Apr 21,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(9.450.75)$
 $= 0.108$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.11, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 114462 \cdot 1.25$
 $= 6730.341$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 114462$
 $= 3589.5$ Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1125	16351.7	959.649



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Earthquake Load Calculation: Step: 15 9:40am Apr 21,2026

20	30	1125	16351.7	959.649
30	40	1128	16351.7	962.209
40	Sad1	1128	16351.7	962.209
Sad1	50	1128	16351.7	962.209
40	50	1128	16351.7	962.209
50	60	1128	16351.7	962.209

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 16 9:40am Apr 21,2026

Shop/Field Installation Options :

Insulation is installed in the Field before being lifted.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4913.000 mm
Center of Gravity of the Liquid	3792.167 mm
Center of Gravity of the Insulation	3524.263 mm
Center of Gravity of the Lining	4755.180 mm
Center of Gravity of the Nozzles	713.046 mm
Center of Gravity of the Added Weights (Operating)	4369.528 mm
Center of Gravity of the Added Weights (Empty)	4369.528 mm
Center of Gravity of the Tubesheet(s)	1670.000 mm
Center of Gravity of the Tubes	4000.000 mm
Center of Gravity of the Baffles	4968.557 mm
Center of Gravity of Bare Shell New and Cold	2434.848 mm
Center of Gravity of Bare Shell Corroded	2393.145 mm
Vessel CG in the Operating Condition	2851.410 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	2592.113 mm
Vessel CG in the Test Condition	2776.885 mm

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Saddle Calcs: Bundle Pull New: Step: 17 9:40am Apr 21, 2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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Saddle Calcs: Bundle Pull Cor: Step: 18 9:40am Apr 21, 2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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Saddle Calcs: Operating Case: Step: 19 9:40am Apr 21,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

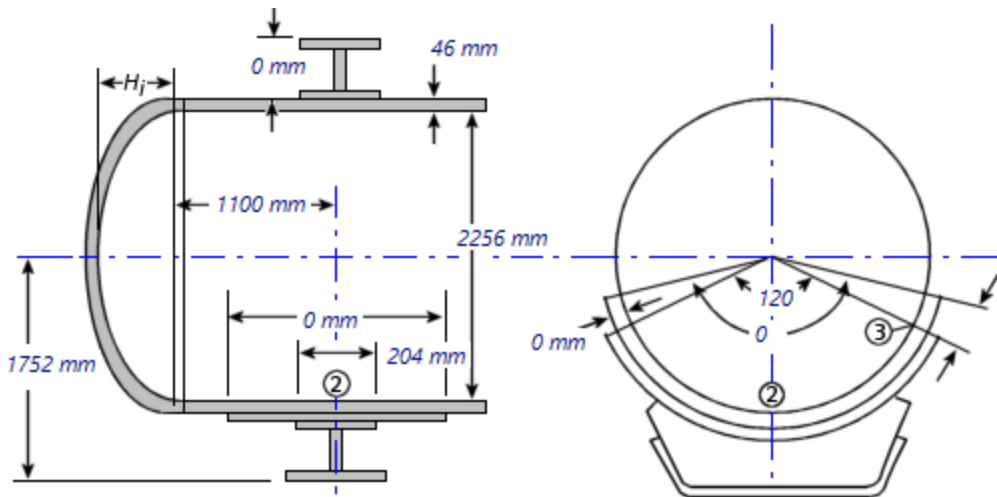
$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{ 1152 \cdot 43 }, 2 \cdot 1100)$$

$$= 551.2796 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1152.50	mm
Shell Thickness used in this Case	t	46.000	mm
Stiffened Vessel Length per 4.15.6	L	5320.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1100.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		132.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1752.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	142220.17	Kgf
Pressure used in Analysis	P	4959.885 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	63.71	137.00
Long. Stress at Bottom of Midspan	69.23	137.00
Long. Stress at Top of Saddles	73.95	137.00
Long. Stress at Bottom of Saddles	62.33	137.00



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Saddle Calcs: Operating Case: Step: 19 9:40am Apr 21,2026

Tangential Shear in Shell	19.32	109.60
Circ. Stress at Horn of Saddle	111.27	171.25
Circ. Compressive Stress in Shell	4.47	137.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (989/2 + 0) \cdot 1752/2061$$

$$= 1260.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(631, 0) \cdot 1752/3200$$

$$= 345.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(6730, 0) \cdot 1752/3200$$

$$= 3684.9 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (6730/2 + 0) \cdot 1752/2061$$

$$= 8581.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 133639 + \max(345, 1261, 3685, 8581)$$

$$= 142220.2 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.55 \cdot 1.2)^2)$$

$$= 6187.588 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	143156.19	Kgf
Transverse Shear Load	3365.17	Kgf
Longitudinal Shear Load	6730.34	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2256/2 + 46 + 16)\sin(120/2), 2 \cdot 1152)$$

$$= 2061.141 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0492	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -142220 \cdot 1100 [1 - (1 - 1100/5320 + (1152^2 - 0^2)/(2 \cdot 1100 \cdot 5320))/(1 + (4 \cdot 0)/(3 \cdot 5320))]$$

$$= -14593.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$



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Saddle Calcs: Operating Case: Step: 19 9:40am Apr 21, 2026

$$= 142220 \cdot 5320 / 4 \left((1 + 2(1152^2 - 0^2) / (5320^2)) / (1 + (4 \cdot 0) / (3 \cdot 5320)) - 4 \cdot 1100 / 5320 \right)$$

$$= 50465.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 4960 \cdot 1152 / (2 \cdot 43) - 50466 / (\pi \cdot 1152^2 \cdot 43)$$

$$= 63.71 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 4960 \cdot 1152 / (2 \cdot 43) + 50466 / (\pi \cdot 1152^2 \cdot 43)$$

$$= 69.23 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4960 \cdot 1152 / (2 \cdot 43) - 14593 / (0.11 \cdot \pi \cdot 1152^2 \cdot 43)$$

$$= 73.95 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4960 \cdot 1152 / (2 \cdot 43) + 14593 / (0.19 \cdot \pi \cdot 1152^2 \cdot 43)$$

$$= 62.33 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 142220 (5320 - 2 \cdot 1100) / (5320 + (4 \cdot 0 / 3))$$

$$= 83407.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 83407 / (1152 \cdot 43)$$

$$= 19.32 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1152 \cdot 43)}$$

$$= 173.640 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 142220 \cdot 0.1 / (43(204 + 174 + 174))$$

$$= -4.47 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -142220 / (4 \cdot 43(204 + 174 + 174)) - 12 \cdot 0.049 \cdot 142220 \cdot 1152 / (5320 \cdot 43^2)$$

$$= -111.27 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(1152 \cdot 43)}, 2 \cdot 1100)$$

$$= 551.28 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 3200.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13139\text{E-}04 \cdot 3200(275 - 21.1)$$

$$= 10.675 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length Bplen 2100.0000 mm



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Saddle Calcs: Operating Case: Step: 19 9:40am Apr 21,2026

Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	440.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194.7	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1131 \cdot 43) + 254})^3) / 12$$

$$= 396.226 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 396 + 257 (265 - 21.5)^2$$

$$= 153176.609 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	598.0	43.0	21.5	257.2	552874.1	0.153E+06
Wearplate	254.0	16.0	51.0	40.6	207264.0	0.187E+05
Web	20.0	517.0	317.5	103.4	3282948.5	0.259E+05
BasePlate	440.0	45.0	598.5	198.0	11850296.0	0.220E+06
Totals	...	...	...	599.2	15893382.0	0.418E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 15893382 / 599$$

$$= 265.247 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 142220$$

$$= 28944.8887 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 8.2990 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116.8100 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1752 - 1131 \cdot \sin(120/2) / 1.05$$

$$= 816.670 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 23638.9023 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 14.7131 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 129.7889 \text{ N./mm}^2$$



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Saddle Calcs: Operating Case: Step: 19 9:40am Apr 21,2026

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(142220 + 936)440}{4 \cdot 2100 \cdot 137}}$$

$$= 40.129 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1128 + 46 + 16)$$

$$= 2061.140 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2061 / (5 - 1)$$

$$= 515.285 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 515 \cdot 440 / 2$$

$$= 1133.627 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 143156 / (2100 \cdot 440)$$

$$= 15.493 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1134 \cdot 15.5$$

$$= 17563.391 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.8 + 51.5$$

$$= 88.329 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 17563 / 88.3$$

$$= 19.500 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	204.0	...	40.8	...	0.141E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 20$$

$$= 184.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 88.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:



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Saddle Calcs: Operating Case: Step: 19 9:40am Apr 21,2026

= Saddle Width / 2
= 204/2
= 102.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1415/88.3}$
= 40.024 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk
= 1752 - $\cos(120/2)$ (1128 + 46 + 16) - 45
= 1112.000 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/195}$
= 142.386

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1112/40
= 27.783

Bending Moment [Rm]:
= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
= $6730 / (2 \cdot 2100) \cdot 515 \cdot 1112/2$
= 459.112 Kg-m.

Compressive Allowable, $KL/r < Cc$ (27.8 < 142) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (27.8)^2 / (2 \cdot 142^2)) 195 / (5/3 + 3 \cdot (27.8) / (8 \cdot 142) - (27.8)^3 / (8 \cdot 142^3))$
= 109.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
= $19.5/110 + (459 \cdot 102/14149440) / 130$
= 0.428

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	184.0	0.0	36.8	0.0	0.141E+04
Web	515.3	20.0	0.0	103.1	0.0	34.4
Totals	...	...	...	139.9	...	0.145E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/140
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 204/2
= 102.000 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 1752 - $\cos(1190/2)$ (45 + 0 + 0) - 0
= 517.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(1448/140)}$$

$$= 32.176 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 517/32.2$$

$$= 16.068$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 6730 / (2 \cdot 2100)$$

$$= 1.602 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.6 \cdot 515 \cdot 517$$

$$= 426.908 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 427/141957$$

$$= 29.492 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($16.1 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (16.1)^2 / (2 \cdot 142^2)) 195 / (5/3 + 3 \cdot (16.1) / (8 \cdot 142) - (16.1)^3 / (8 \cdot 142^3))$$

$$= 113.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 18.9/113 + (427 \cdot 102/1448) / 130$$

$$= 0.394$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	129.12 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	134574.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	3684.9 Kg
Maximum Transverse Force	Ft	3365.2 Kg
Maximum Longitudinal Force	F1	6730.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6730 / (4.87 \cdot 8)$$

$$= 17.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3365 / (4.87 \cdot 8)$$

$$= 8.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $6730 / (82.8 \cdot 8)$
= 0.9964 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1752 \cdot 3365 + 0$
= 5895.90 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $5896 / 134575$
= 43.81 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 1, 0]$
= 0.9964 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 16$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.96$, Temp. Reduction = 2 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-33 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
= $\min(204 + 1.56 \cdot \sqrt{1152 \cdot 43}, 2 \cdot 1100)$
= 551.2796 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1152.50	mm
Shell Thickness used in this Case	t	46.000	mm
Stiffened Vessel Length per 4.15.6	L	5320.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1100.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	588.45	mm
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	



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Circumferential Efficiency at Mid-Span		1.00
Distance from Saddle Base to Centerline	B	1752.00 mm
Coefficient of Friction	mu	0.40

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		25886.58 Kgf
Pressure used in Analysis	P	4959.885 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	66.27	137.00
Long. Stress at Bottom of Midspan	66.67	137.00
Long. Stress at Top of Saddles	69.91	137.00
Long. Stress at Bottom of Saddles	64.57	137.00
Tangential Shear in Shell	3.07	109.60
Circ. Stress at Horn of Saddle	20.25	171.25
Circ. Compressive Stress in Shell	0.81	137.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (989/2 + 0) \cdot 1752/2061$
 $= 1260.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(631, 0) \cdot 1752/3200$
 $= 345.5 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(6730, 0) \cdot 1752/3200$
 $= 3684.9 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (6730/2 + 0) \cdot 1752/2061$
 $= 8581.3 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 17305 + \max(345, 1261, 3685, 8581)$
 $= 25886.6 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (2.55 \cdot 1.2)^2)$
 $= 6187.588 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	26822.60 Kgf
Transverse Shear Load	3365.17 Kgf
Longitudinal Shear Load	6730.34 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2 (ShellID/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2 (2256/2 + 46 + 16) \sin(120/2), 2 \cdot 1152)$



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= 2061.141 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0492	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -25887 \cdot 1100 \left[1 - \left(1 - \frac{1100}{5320} + \frac{(1152^2 - 588^2)}{(2 \cdot 1100 \cdot 5320)} \right) / \left(1 + \frac{(4 \cdot 588)}{(3 \cdot 5320)} \right) \right]$$

$$= -6708.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{1 + 2(Rm^2 - hm^2)/(L^2)}{1 + (4hm^2)/(3L)} - 4a/L \right)$$

$$= 25887 \cdot 5320 / 4 \left(\frac{1 + 2(1152^2 - 588^2)/(5320^2)}{1 + (4 \cdot 588)/(3 \cdot 5320)} - 4 \cdot 1100 / 5320 \right)$$

$$= 3611.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 4960 \cdot 1152 / (2 \cdot 43) - 3611 / (\pi \cdot 1152^2 \cdot 43)$$

$$= 66.27 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 4960 \cdot 1152 / (2 \cdot 43) + 3611 / (\pi \cdot 1152^2 \cdot 43)$$

$$= 66.67 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 4960 \cdot 1152 / (2 \cdot 43) - 6709 / (0.11 \cdot \pi \cdot 1152^2 \cdot 43)$$

$$= 69.91 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 4960 \cdot 1152 / (2 \cdot 43) + 6709 / (0.19 \cdot \pi \cdot 1152^2 \cdot 43)$$

$$= 64.57 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 25887 (5320 - 2 \cdot 1100) / (5320 + (4 \cdot 588 / 3))$$

$$= 13230.4 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 13230 / (1152 \cdot 43)$$

$$= 3.07 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{1152 \cdot 43}$$

$$= 173.640 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 25887 \cdot 0.1 / (43(204 + 174 + 174))$$

$$= -0.81 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -25887 / (4 \cdot 43(204 + 174 + 174)) - 12 \cdot 0.049 \cdot 25887 \cdot 1152 / (5320 \cdot 43^2)$$



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$$= -20.25 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(1152 \cdot 43)}, 2 \cdot 1100)$$

$$= 551.28 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2100.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	440.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194.7	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1131 \cdot 43)} + 254)43^3)/12$$

$$= 396.226 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 396 + 257 (265 - 21.5)^2$$

$$= 153176.609 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	598.0	43.0	21.5	257.2	552874.1	0.153E+06
Wearplate	254.0	16.0	51.0	40.6	207264.0	0.187E+05
Web	20.0	517.0	317.5	103.4	3282948.5	0.259E+05
BasePlate	440.0	45.0	598.5	198.0	11850296.0	0.220E+06
Totals	...	...	...	599.2	15893382.0	0.418E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 15893382/599$$

$$= 265.247 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 25887$$

$$= 5268.4795 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.5106 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116.8100 \text{ N./mm}^2$$



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Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1752 - 1131 \cdot \sin(120/2) / 1.05$$

$$= 816.670 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 4302.6963 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 2.6780 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 129.7889 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(25887 + 936)440 / (4 \cdot 2100 \cdot 137)}$$

$$= 17.370 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1128 + 46 + 16)$$

$$= 2061.140 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2061 / (5 - 1)$$

$$= 515.285 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 515 \cdot 440 / 2$$

$$= 1133.627 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 26823 / (2100 \cdot 440)$$

$$= 2.903 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1134 \cdot 2.9$$

$$= 3290.782 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.8 + 51.5$$

$$= 88.329 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 3291 / 88.3$$

$$= 3.654 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	204.0	...	40.8	...	0.141E+04

Rib dimension [D]:



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= Saddle Width - Web Thickness

= 204 - 20

= 184.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/88.3

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 204/2

= 102.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1415/88.3}$

= 40.024 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1752 - cos(120/2)(1128 + 46 + 16) - 45

= 1112.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/195}$

= 142.386

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1112/40

= 27.783

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 6730 / (2 • 2100) • 515 • 1112/2

= 459.112 Kg-m.

Compressive Allowable, KL/r < Cc (27.8 < 142) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (27.8)^2 / (2 \cdot 142^2)) 195 /$

$(5/3 + 3 \cdot (27.8) / (8 \cdot 142) - (27.8^3) / (8 \cdot 142^3))$

= 109.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 3.65/110 + (459 • 102/14149440) / 130

= 0.283

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	184.0	0.0	36.8	0.0	0.141E+04
Web	515.3	20.0	0.0	103.1	0.0	34.4
Totals	...	...	...	139.9	...	0.145E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/140

= 0.000 mm

Distance to Centroid [C1]:



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= Saddle Width / 2
= 204/2
= 102.000 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 1752 - cos(1190/2)(45 + 0 + 0) - 0
= 517.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1448/140}$
= 32.176 mm

Slenderness ratio [KL/r]:
= KL/r
= 1 • 517/32.2
= 16.068

Unit Force [Force,u]:
= FI / (2 • Baseplate Length)
= 6730/(2 • 2100)
= 1.602 Kg/mm

Moment at base of inner Rib [Mbase,c]:
= Unit Force • e • L
= 1.6 • 515 • 517
= 426.908 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
= Bending Moment / Section Modulus
= 427/141957
= 29.492 N./mm²

Compressive Allowable, KL/r < Cc (16.1 < 142) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (16.1)^2 / (2 \cdot 142^2)) 195 / (5/3 + 3 \cdot (16.1) / (8 \cdot 142) - (16.1)^3 / (8 \cdot 142^3))$
= 113.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
= Sc/Sca + (Mbase,c • C1/I)/Sba
= 3.55/113 + (427 • 102/1448)/130
= 0.259

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	129.12 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	18241.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	3684.9 Kg
Maximum Transverse Force	Ft	3365.2 Kg
Maximum Longitudinal Force	F1	6730.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
= FI / (Bolt Area • Number of Bolts)



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Saddle Calcs: Operating Case: Step: 19 9:40am Apr 21,2026

= $6730 / (4.87 \cdot 8)$
 = 17.0 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $3365 / (4.87 \cdot 8)$
 = 8.5 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $6730 / (82.8 \cdot 8)$
 = 0.9964 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
 = $1752 \cdot 3365 + 0$
 = 5895.90 Kg-m.

Eccentricity (e):

= R_{mom} / QO
 = $5896 / 18241$
 = 323.21 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 1, 0]$
 = 0.9964 cm²

MDMT Calculations for: Left saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.96$, Temp. Reduction = 2 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

Step: 20 9:40am Apr 21,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

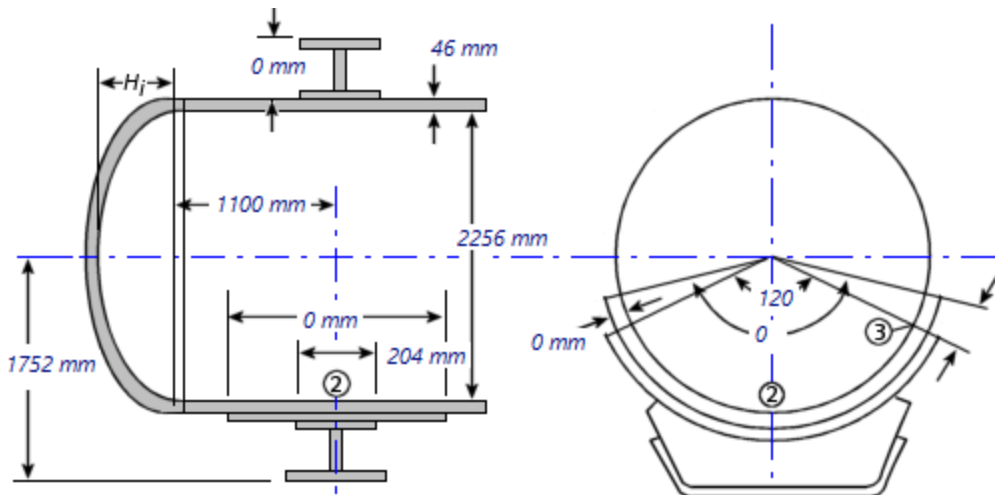
$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{ 1151 \cdot 46 }, 2 \cdot 1100)$$

$$= 562.9559 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1151.00	mm
Shell Thickness used in this Case	t	46.000	mm
Stiffened Vessel Length per 4.15.6	L	5320.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1100.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1752.00	mm
Coefficient of Friction	mu	0.40	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		19947.65	Kgf
Pressure used in Analysis	P	6460.434	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	80.47	234.00
Long. Stress at Bottom of Midspan	81.19	234.00
Long. Stress at Top of Saddles	81.82	234.00
Long. Stress at Bottom of Saddles	80.28	234.00



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Tangential Shear in Shell	2.54	138.58
Circ. Stress at Horn of Saddle	13.73	350.99
Circ. Compressive Stress in Shell	0.57	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (494/2 + 0) \cdot 1752/2061$
 $= 630.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(315, 0) \cdot 1752/3200$
 $= 172.7 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 19317 + \max(173, 630, 0, 0)$
 $= 19947.7 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (2.55 \cdot 1.2)^2)$
 $= 6187.588 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	20883.67	Kgf
Transverse Shear Load	247.19	Kgf
Longitudinal Shear Load	315.48	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2256/2 + 46 + 16) \sin(120/2), 2 \cdot 1151)$
 $= 2061.141 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0493	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -19948 \cdot 1100 [1 - (1 - 1100/5320 + (1151^2 - 0^2)/(2 \cdot 1100 \cdot 5320)) / (1 + (4 \cdot 0)/(3 \cdot 5320))]$
 $= -2053.3 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 19948 \cdot 5320/4 ((1 + 2(1151^2 - 0^2)/(5320^2)) / (1 + (4 \cdot 0)/(3 \cdot 5320)) - 4 \cdot 1100/5320)$
 $= 7071.8 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 6460 \cdot 1151/(2 \cdot 46) - 7072/(\pi \cdot 1151^2 \cdot 46)$
 $= 80.47 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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Saddle Calcs: Test Case: Step: 20 9:40am Apr 21,2026

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 6460 \cdot 1151 / (2 \cdot 46) + 7072 / (\pi \cdot 1151^2 \cdot 46) \\
 &= 81.19 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 6460 \cdot 1151 / (2 \cdot 46) - 2053 / (0.11 \cdot \pi \cdot 1151^2 \cdot 46) \\
 &= 81.82 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 6460 \cdot 1151 / (2 \cdot 46) + 2053 / (0.19 \cdot \pi \cdot 1151^2 \cdot 46) \\
 &= 80.28 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot hm / 3)) \\
 &= 19948 (5320 - 2 \cdot 1100) / (5320 + (4 \cdot 0 / 3)) \\
 &= 11698.6 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 11699 / (1151 \cdot 46) \\
 &= 2.54 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1151 \cdot 46)} \\
 &= 179.478 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.76 \cdot 19948 \cdot 0.1 / (46(204 + 179 + 179)) \\
 &= -0.57 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ^*7]:

$$\begin{aligned}
 &= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2) \\
 &= -19948 / (4 \cdot 46(204 + 179 + 179)) - \\
 &\quad 12 \cdot 0.049 \cdot 19948 \cdot 1151 / (5320 \cdot 46^2) \\
 &= -13.73 \text{ N./mm}^2
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(204 + 1.56 \cdot \sqrt{(1151 \cdot 46)}, 2 \cdot 1100) \\
 &= 562.96 \text{ mm}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2100.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	440.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(1128 \cdot 46)} + 254)^3) / 12 \\
 &= 494.265 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:



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$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 494 + 280 (259 - 23)^2 \\
 &= 155956.250 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	609.4	46.0	23.0	280.3	644693.7	0.156E+06
Wearplate	254.0	16.0	54.0	40.6	219456.0	0.170E+05
Web	20.0	517.0	320.5	103.4	3313970.5	0.270E+05
BasePlate	440.0	45.0	601.5	198.0	11909700.0	0.233E+06
Totals	...	...	...	622.3	16087821.0	0.433E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 16087821/622 \\
 &= 258.505 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 19948 \\
 &= 4059.7793 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 1.1640 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 1752 - 1128 \cdot \sin(120/2) / 1.05 \\
 &= 819.152 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3325.6404 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.9460 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(19948 + 936)440) / (4 \cdot 2100 \cdot 173)} \\
 &= 13.626 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (1128 + 46 + 16) \\
 &= 2061.140 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 2061 / (5 - 1) \\
 &= 515.285 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



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$$= 515 \cdot 440/2$$

$$= 1133.627 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 20884 / (2100 \cdot 440)$$

$$= 2.260 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1134 \cdot 2.26$$

$$= 2562.153 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.8 + 51.5$$

$$= 88.329 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2562 / 88.3$$

$$= 2.845 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	204.0	...	40.8	...	0.141E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 20$$

$$= 184.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/88.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1415/88.3)}$$

$$= 40.024 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1752 - \cos(120/2) (1128 + 46 + 16) - 45$$

$$= 1112.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1112/40$$

$$= 27.783$$



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Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 315 / (2 \cdot 2100) \cdot 515 \cdot 1112 / 2$$

$$= 21.521 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($27.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (27.8)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (27.8) / (8 \cdot 123) - (27.8^3) / (8 \cdot 123^3))$$

$$= 144.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.84 / 145 + (21.5 \cdot 102 / 14149440) / 173$$

$$= 0.028$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	184.0	0.0	36.8	0.0	0.141E+04
Web	515.3	20.0	0.0	103.1	0.0	34.4
Totals	...	...	...	139.9	...	0.145E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 140$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204 / 2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1752 - \cos(1190 / 2) (45 + 0 + 0) - 0$$

$$= 517.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1448 / 140) }$$

$$= 32.176 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 517 / 32.2$$

$$= 16.068$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 315 / (2 \cdot 2100)$$

$$= 0.075 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.075 \cdot 515 \cdot 517$$

$$= 20.011 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 20 / 141957$$

$$= 1.382 \text{ N./mm}^2$$



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Compressive Allowable, $KL/r < C_c$ ($16.1 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (16.1)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (16.1) / (8 \cdot 123) - (16.1^3) / (8 \cdot 123^3))$$

$$= 150.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 2.76 / 150 + (20 \cdot 102 / 1448) / 173$$

$$= 0.026$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	20253.3 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	172.7 Kgf
Maximum Transverse Force	Ft	247.2 Kgf
Maximum Longitudinal Force	F1	315.5 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 315 / (4.87 \cdot 8)$$

$$= 0.8 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 247 / (4.87 \cdot 8)$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 315 / (29 \cdot 8)$$

$$= 0.1335 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1752 \cdot 247 + 0$$

$$= 433.08 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 433 / 20253$$

$$= 21.38 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:



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Saddle Calcs: Test Case: Step: 20 9:40am Apr 21,2026

= max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.13, 0]
 = 0.1335 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 = $\min(204 + 1.56 \cdot \sqrt{1151 \cdot 46}, 2 \cdot 1100)$
 = 562.9559 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1151.00	mm
Shell Thickness used in this Case	t	46.000	mm
Stiffened Vessel Length per 4.15.6	L	5320.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1100.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	586.95	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1752.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		19943.52	Kgf
Pressure used in Analysis	P	6460.434	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	80.69	234.00
Long. Stress at Bottom of Midspan	80.97	234.00
Long. Stress at Top of Saddles	83.31	234.00
Long. Stress at Bottom of Saddles	79.45	234.00
Tangential Shear in Shell	2.21	138.58
Circ. Stress at Horn of Saddle	13.73	350.99
Circ. Compressive Stress in Shell	0.57	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 = $3 (494/2 + 0) \cdot 1752/2061$
 = 630.3 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= $\max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 = $\max(315, 0) \cdot 1752/3200$
 = 172.7 Kgf



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Saddle Calcs: Test Case: Step: 20 9:40am Apr 21,2026

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 19313 + max(173, 630, 0, 0)

= 19943.5 Kgf

Longitudinal Wind Force [F_L]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766 (0 + $\pi/4$ (2.55 • 1.2)²)

= 6187.588 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 20879.54 Kgf

Transverse Shear Load 247.19 Kgf

Longitudinal Shear Load 315.48 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(2256/2 + 46 + 16)sin(120/2), 2•1151)

= 2061.141 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0493 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -19944•1100[1-(1-1100/5320+(1151²-587²)/

(2•1100•5320))/(1+(4•587)/(3•5320))]

= -5165.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 19944•5320/4((1+2(1151²-587²)/(5320²))/(1+(4•587)/

(3•5320))-4•1100/5320)

= 2787.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 6460 • 1151/(2•46) - 2787/(π •1151²•46)

= 80.69 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 6460 • 1151/(2 • 46) + 2787/(π • 1151² • 46)

= 80.97 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)

= 6460•1151/(2•46)--5166/(0.11• π •1151²•46)

= 83.31 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 6460•1151/(2•46)+5166/(0.19• π •1151²•46)

= 79.45 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 19944 (5320 - 2 • 1100)/(5320 + (4 • 587/3))



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Saddle Calcs: Test Case: Step: 20 9:40am Apr 21,2026

= 10196.3 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 10196 / (1151 \cdot 46)$$

$$= 2.21 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1151 \cdot 46)}$$

$$= 179.478 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 19944 \cdot 0.1 / (46(204 + 179 + 179))$$

$$= -0.57 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -19944 / (4 \cdot 46(204 + 179 + 179)) -$$

$$12 \cdot 0.049 \cdot 19944 \cdot 1151 / (5320 \cdot 46^2)$$

$$= -13.73 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(1151 \cdot 46)}, 2 \cdot 1100)$$

$$= 562.96 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2100.0000	mm
Baseplate Thickness	Bpthk	45.0000	mm
Baseplate Width	Bpwid	440.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1128 \cdot 46)} + 254)^3) / 12$$

$$= 494.265 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 494 + 280(259 - 23)^2$$

$$= 155956.250 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	609.4	46.0	23.0	280.3	644693.7	0.156E+06
Wearplate	254.0	16.0	54.0	40.6	219456.0	0.170E+05
Web	20.0	517.0	320.5	103.4	3313970.5	0.270E+05
BasePlate	440.0	45.0	601.5	198.0	11909700.0	0.233E+06
Totals	...	...	...	622.3	16087821.0	0.433E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 16087821 / 622$$



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Saddle Calcs: Test Case: Step: 20 9:40am Apr 21,2026

$$= 258.505 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 19944$$

$$= 4058.9380 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.1638 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1752 - 1128 \cdot \sin(120/2) / 1.05$$

$$= 819.152 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3324.9512 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.9456 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(19944 + 936) \cdot 440) / (4 \cdot 2100 \cdot 173)}$$

$$= 13.625 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1128 + 46 + 16)$$

$$= 2061.140 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 2061 / (5 - 1)$$

$$= 515.285 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 515 \cdot 440 / 2$$

$$= 1133.627 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 20880 / (2100 \cdot 440)$$

$$= 2.260 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1134 \cdot 2.26$$

$$= 2561.646 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.8 + 51.5$$



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Saddle Calcs: Test Case: Step: 20 9:40am Apr 21,2026

$$= 88.329 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 2562/88.3$$

$$= 2.844 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	204.0	...	40.8	...	0.141E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 20$$

$$= 184.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/88.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1415/88.3)}$$

$$= 40.024 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1752 - \cos(120/2) (1128 + 46 + 16) - 45$$

$$= 1112.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1112/40$$

$$= 27.783$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 315 / (2 \cdot 2100) \cdot 515 \cdot 1112/2$$

$$= 21.521 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($27.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (27.8)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (27.8) / (8 \cdot 123) - (27.8^3) / (8 \cdot 123^3))$$

$$= 144.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.84/145 + (21.5 \cdot 102/14149440) / 173$$

$$= 0.028$$



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Saddle Calcs: Test Case:

Step: 20 9:40am Apr 21,2026

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	184.0	0.0	36.8	0.0	0.141E+04
Web	515.3	20.0	0.0	103.1	0.0	34.4
Totals	...	...	...	139.9	...	0.145E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/140$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1752 - \cos(1190/2)(45 + 0 + 0) - 0$$

$$= 517.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1448/140)}$$

$$= 32.176 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 517/32.2$$

$$= 16.068$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 315/(2 \cdot 2100)$$

$$= 0.075 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.075 \cdot 515 \cdot 517$$

$$= 20.011 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 20/141957$$

$$= 1.382 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($16.1 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (16.1)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (16.1) / (8 \cdot 123) - (16.1^3) / (8 \cdot 123^3))$$

$$= 150.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 2.76/150 + (20 \cdot 102/1448) / 173$$

$$= 0.026$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate

Nbolts

8

Total Number of Bolts in Tension/Baseplate

Nbt

4



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Saddle Calcs: Test Case: Step: 20 9:40am Apr 21,2026

Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load QO (Weight)	QO	20249.2 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	172.7 Kgf
Maximum Transverse Force	Ft	247.2 Kgf
Maximum Longitudinal Force	F1	315.5 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 315 / (4.87 \cdot 8)$$

$$= 0.8 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 247 / (4.87 \cdot 8)$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 315 / (29 \cdot 8)$$

$$= 0.1335 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1752 \cdot 247 + 0$$

$$= 433.08 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 433 / 20249$$

$$= 21.39 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.13, 0]$$

$$= 0.1335 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: Manhole

Nozl: 13 9:40am Apr 21,2026

Input, Nozzle Desc: Manhole

From: 10

Pressure for Reinforcement Calculations	P	17143.129	KPa.
Temperature for Internal Pressure	Temp	250	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2250.00	mm
Head Finished (Minimum) Thickness	t	73.9500	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	47.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	88.3346	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	62.9000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



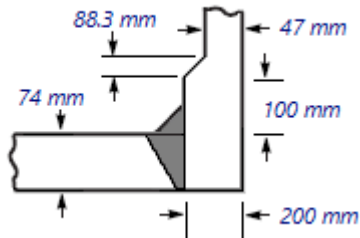
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Nozzle Calcs.: Manhole

Noz1: 13 9:40am Apr 21,2026



Reinforcement CALCULATION, Description: Manhole

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.850 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 17143 \cdot 1128 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 17143)$
 $= 70.9500 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 17143 \cdot 308 / (136 \cdot 1 - 0.6 \cdot 17143)$
 $= 41.9766 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.9600 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 153 / 0.5773$
 $= 265.0039 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001	mm
Parallel to Vessel Wall, opening length	d	615.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	177.3750	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.986$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.986$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$



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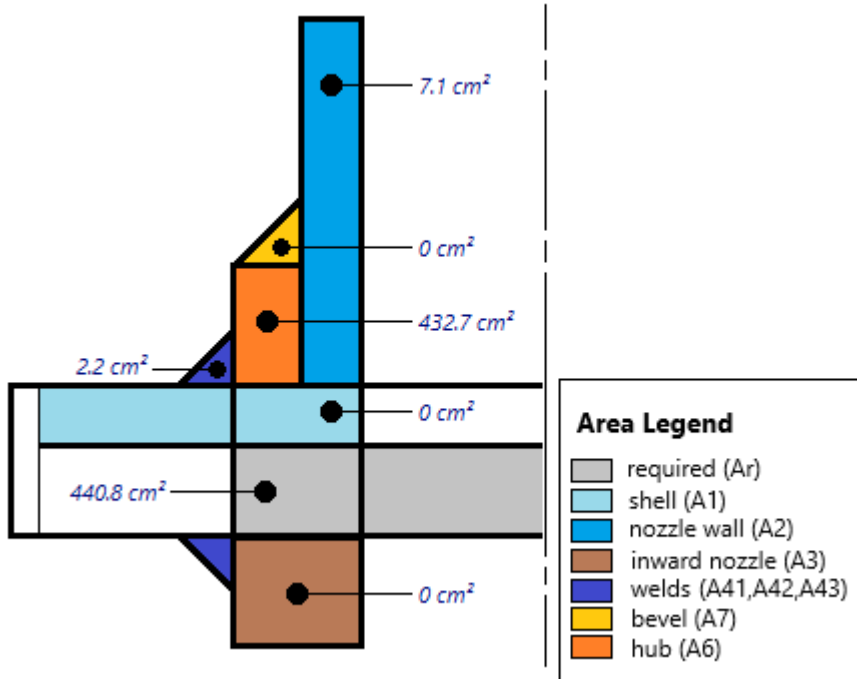
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Nozzle Calcs.: Manhole

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= min(0.99, 1)
= 0.986



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	440.818	10.777	NA
Area in Shell	A1	0.000	411.560	NA
Area in Nozzle Wall	A2	7.074	146.977	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.217	2.217	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	432.711	432.711	NA
TOTAL AREA AVAILABLE	Atot	442.002	993.465	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)
= (616 • 71 • 1 + 2 • 197 • 71 • 1 • (1-0.99))
= 440.818 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)
= 616 (1 • 71 - 1 • 71) - 2 • 197
(1 • 71 - 1 • 71) • (1 - 0.99)
= 0.000 cm²

Area Available in Nozzle Projecting Outward [A2]:
= (2 • tlnp) (tn - trn) fr2



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Nozzle Calcs.: Manhole

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$$= (2 \cdot 177)(44 - 42)0.99$$

$$= 7.074 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 0.99 + (0)^2 \cdot 0.99$$

$$= 2.217 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(177, 500, 100)) \cdot (200 - 47) \cdot 0.99$$

$$= 432.711 \text{ cm}^2 \text{ Includes Bevel Area of } 131.144 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 44.9766 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 73.9500 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 73.9500 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.4141 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(74, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(45, 11.3)$$

$$= 44.9766 \text{ mm}$$

Available Nozzle Neck Thickness = 47.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 47, tr = 41.3, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.94, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	32 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 74, tr = 71, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-36 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 17143.13/25510.00 = 0.672$$

Weld Size Calculations, Description: Manhole

Intermediate Calc. for nozzle/shell welds	Tmin	19.0000 mm
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Nozzle Calcs.: Manhole

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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (441 - 0 + 2 \cdot 197 \cdot 0.99 \cdot (1 \cdot 71 - 71)) \cdot 138) \\
 &= 620300.00 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (440 + 2.22 - 0 \cdot 0.99) \cdot 138 \\
 &= 621966.62 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (440 + 0 + 2.22 + (275)) \cdot 138 \\
 &= 1009629.06 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (440 + 0 + 2.22 + 0 + (275)) \cdot 138 \\
 &= 1009629.06 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 1010 \cdot 15 \cdot 0.49 \cdot 136 \\
 &= 161645. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 406) \cdot (200 - 3) \cdot 0.7 \cdot 136 \\
 &= 2440990. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{g_{nvi}} \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 1010 \cdot (62.9 - 0) \cdot 0.74 \cdot 138 \\
 &= 1038710. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (161645 + 2440990) = 2602635 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (161645 + 0 + 1038709 + 0) = 1200354 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (161645 + 1038709 + 0) = 1200354 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 2602635 Kgf, must exceed W = 620300 Kgf or W1 = 621966 Kgf



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Nozzle Calcs.: Manhole

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Path 2-2 = 1200354 Kgf, must exceed W = 620300 Kgf or W2 = 1009629 Kgf

Path 3-3 = 1200354 Kgf, must exceed W = 620300 Kgf or W3 = 1009629 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 17143 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 119.6135 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 693.5635 mm

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Nozzle Calcs.: N1- Tube inlet

Nozl: 14 9:40am Apr 21,2026

Input, Nozzle Desc: N1- Tube inlet

From: 20

Pressure for Reinforcement Calculations	P	17163.820	KPa.
Temperature for Internal Pressure	Temp	250	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2250.00	mm
Design Length of Section	L	1875.0000	mm
Shell Finished (Minimum) Thickness	t	155.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		750.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	130.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		250.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	155.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



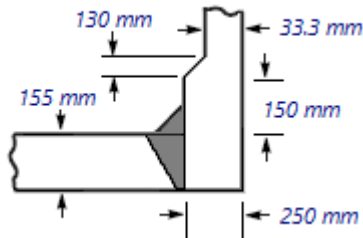
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Nozzle Calcs.: N1- Tube inlet

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Reinforcement CALCULATION, Description: N1- Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	10.126 in.
Actual Thickness Used in Calculation	1.312 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 17164 \cdot 1128 / (138 \cdot 1 - 0.6 \cdot 17164)$$

$$= 151.6220 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 17164 \cdot 132 / (136 \cdot 1 - 0.6 \cdot 17164)$$

$$= 17.9706 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.2507 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 152 / 0.5773$$

$$= 263.2718 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	627.8499	mm
Parallel to Vessel Wall	Rn+tn+t	313.9250	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	339.0838	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 136 / 138)$$

$$= 0.986$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 136 / 138)$$

$$= 0.986$$



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Nozzle Calcs.: N1- Tube inlet

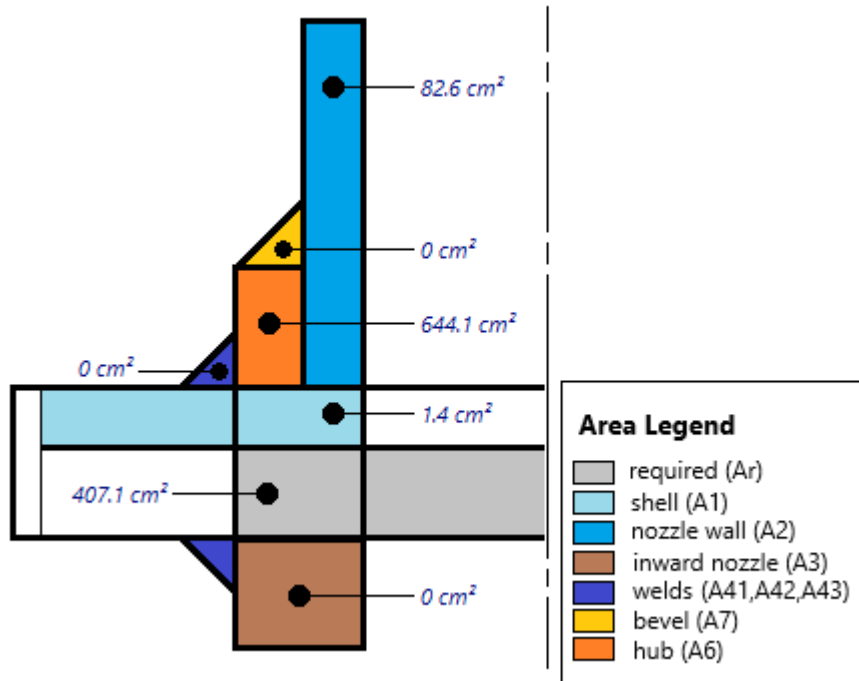
Nozl: 14 9:40am Apr 21,2026

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.99, 1)

= 0.986



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	407.079	9.692	NA
Area in Shell	A1	1.359	520.292	NA
Area in Nozzle Wall	A2	82.569	194.315	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	644.131	644.131	NA
TOTAL AREA AVAILABLE	Atot	728.059	1358.738	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (263•152•1+2•182•152•1•(1-0.99))

= 407.079 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)

= 365 (1 • 152 - 1 • 152) - 2 • 182

(1 • 152 - 1 • 152) • (1 - 0.99)

= 1.359 cm²



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 9:40am Apr 21,2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 339) (30.3 - 18) 0.99$$

$$= 82.569 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_{r2} + (W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot f_{r2}$$

$$= (20^2 - 4) \cdot 0.99 + (0^2 - 0) \cdot 0.99$$

$$= 0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(339, 500, 150)) \cdot (185 - 33.3) \cdot 0.99$$

$$= 644.131 \text{ cm}^2 \text{ Includes Bevel Area of } 194.737 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 20.9706 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 5.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 154.6220 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 154.6220 \text{ mm}$
Wall Thickness, shell/head, external pressure	$tr_{b2} = 3.8284 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(155, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(21, 11.3)$$

$$= 20.9706 \text{ mm}$$

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

$$= \max(t_{UG-45}, tr_{B31.3})$$

$$= \max(21, 15.5)$$

$$= 20.9706 \text{ mm}$$

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = $0.875 \cdot 33.325 = 29.159 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 52.4,	Allowable	: 136.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 290.1 N./mm ²	Passed
Occasional	: 33.4,	Allowable	: 180.9 N./mm ²	Passed
Shear	: 12.9,	Allowable	: 95.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 29.2$, $t_r = 17.7$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.68$, Temp. Reduction = $18 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	23 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	5 $^\circ\text{C}$



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 9:40am Apr 21,2026

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C

Weld Size Calculations, Description: N1- Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (407 - 1.36 + 2 \cdot 182 \cdot 0.99 \cdot (1 \cdot 152 - 152)) \cdot 138) \\
 &= 572824.12 \text{ Kg}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can}) / 707)^2 \cdot f_r2 \cdot S_v \\
 &= (727 + 0.34108E-05 - 0 \cdot 0.99) \cdot 138 \\
 &= 1022581.06 \text{ Kg}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v \\
 &= (727 + 0 + 0.34108E-05 + (546)) \cdot 138 \\
 &= 1791223.62 \text{ Kg}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\
 &= (727 + 0 + 0.34108E-05 + 0 + (546)) \cdot 138 \\
 &= 1791223.62 \text{ Kg}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 628 \cdot 20 \cdot 0.49 \cdot 136 \\
 &= 134032. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 223) \cdot (185 - 3) \cdot 0.7 \cdot 136 \\
 &= 1238628. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 628 \cdot (155 - 3) \cdot 0.74 \cdot 138 \\
 &= 1560967. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



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Nozzle Calcs.: N1- Tube inlet

Nozl: 14 9:40am Apr 21,2026

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (134032 + 1238628) = 1372659 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (134032 + 0 + 1560966 + 0) = 1694998 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (134032 + 1560966 + 0) = 1694998 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1372659 Kgf, must exceed W = 572824 Kgf or W1 = 1022581 Kgf
 Path 2-2 = 1694998 Kgf, must exceed W = 572824 Kgf or W2 = 1791223 Kgf
 Path 3-3 = 1694998 Kgf, must exceed W = 572824 Kgf or W3 = 1791223 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 17183 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 44.6870 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 699.6870 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 313.92 mm

Input Echo, WRC297 Item 1, Description: N1- Tube inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2250.000	mm
Shell Thickness	155.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	257.200	mm
Nozzle Thickness	250.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 136.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	17163.82	KPa.
Radial Load	P	-3823.00	Kgf
Circumferential Shear	Vc	3058.00	Kgf



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 9:40am Apr 21,2026

Longitudinal Shear	Vl	3058.00	Kgf
Circumferential Moment	Mc	2753.01	Kg-m.
Longitudinal Moment	Ml	2753.01	Kg-m.
Torsional Moment	Mt	3211.01	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, D_v/T_v (16.516) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_v (4.885) is < 5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_n (3.029) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 757.2 + 2 \cdot 1.65 \cdot \sqrt{1204.0(155.0 - 3.0)}$$

$$= 2168.922 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2408.000 mm
Nozzle Outside Diameter	(d) =	757.200 mm
Vessel Thickness used	(T) =	152.000 mm
Nozzle Thickness used	(t) =	247.000 mm
T / t	=	0.615
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.252$$

Nr/P	=	0.097
Mr/P	=	0.187
M0/P	=	0.056
N0/P	=	0.120
MrD/Mc	=	0.296
NrDL/Mc	=	0.107
M0d/Mc	=	0.096
N0DL/Mc	=	0.048
MrD/Ml	=	0.168
NrDL/Ml	=	0.087
M0D/Ml	=	0.056
N0DL/Ml	=	0.130

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	119	136	119	136
Outplane Stress Summary	119	136	120	135

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



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Nozzle Calcs.: N1- Tube inlet

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longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	59	59	63	56

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top	Top	Bottom	Bottom
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		0	0	0	0
Shear Stress Summary		0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities		Top	Top	Bottom	Bottom
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		119	136	120	135

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Membrane (P)		0	0	0	0
Outplane Bending (P)		0	0	0	0
Outplane Membrane (Mc)		0	0	0	0
Outplane Bending (Mc)		0	0	0	0
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		59	59	59	59
Outplane Stress Summary		59	60	61	58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Inplane Membrane (P)		0	0	0	0
Inplane Bending (P)		1	-1	1	-1
Inplane Membrane (Mc)		0	0	0	0
Inplane Bending (Mc)		-2	2	2	-2
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		119	136	119	136
Inplane Stress Summary		118	137	124	132

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 9:40am Apr 21,2026

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	118	137	124	132
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	26	5	26

Hoop Stress Summary	5	26	6	26
---------------------	---	----	---	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	2	2	4	2
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	5	26	6	26
------------------------	---	----	---	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
--------------------	---	---	---	---



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Nozzle Calcs.: N1- Tube inlet Nozl: 14 9:40am Apr 21,2026

Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	26	5	26
Hoop Stress Summary	5	26	5	26

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	1	2	4	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	5	26	5	26

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	119	136	119	136	119	136	119	136
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	-1	1	3	-3
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	2	-2	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	119.0	136.0	119.0	136.0	119.0	136.0	119.0	136.0
Pm+Pl (SUS)	119.0	136.0	119.0	136.0	119.0	136.0	119.0	136.0
Pm+Pl+Q (Total)	119.0	136.0	119.0	136.0	118.0	137.0	122.0	133.0



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Nozzle Calcs.: N1- Tube inlet

Nozl: 14

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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	136.00	207.00	Passed
Pm+Pl+Q (TOTAL)	137.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	26	5	26	5	26	5	26
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	5.0	26.0	5.0	26.0	5.0	26.0	5.0	26.0
Pm+Pl (SUS)	5.0	26.0	5.0	26.0	5.0	26.0	5.0	26.0
Pm+Pl+Q (Total)	5.0	26.0	5.0	26.0	5.0	26.0	5.0	26.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	136.00	Passed
Pm+Pl (SUS)	26.00	204.00	Passed
Pm+Pl+Q (TOTAL)	26.00	408.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N1- inlet

Nozl: 15 9:40am Apr 21,2026

Input, Nozzle Desc: N1- inlet

From: 20

Pressure for Reinforcement Calculations	P	17163.820	KPa.
Temperature for Internal Pressure	Temp	250	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2250.00	mm
Design Length of Section	L	1875.0000	mm
Shell Finished (Minimum) Thickness	t	155.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		750.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	130.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		250.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	155.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



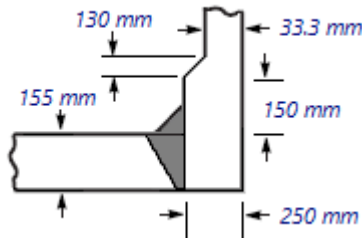
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Nozzle Calcs.: N1- inlet

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Reinforcement CALCULATION, Description: N1- inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	10.126 in.
Actual Thickness Used in Calculation	1.312 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 17164 \cdot 1128 / (138 \cdot 1 - 0.6 \cdot 17164)$
 $= 151.6220$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 17164 \cdot 132 / (136 \cdot 1 - 0.6 \cdot 17164)$
 $= 17.9706$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.2507 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 152 / 0.5773$
 $= 263.2718$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	627.8499	mm
Parallel to Vessel Wall	Rn+tn+t	313.9250	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	339.0838	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.986$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.986$



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Nozzle Calcs.: N1- inlet

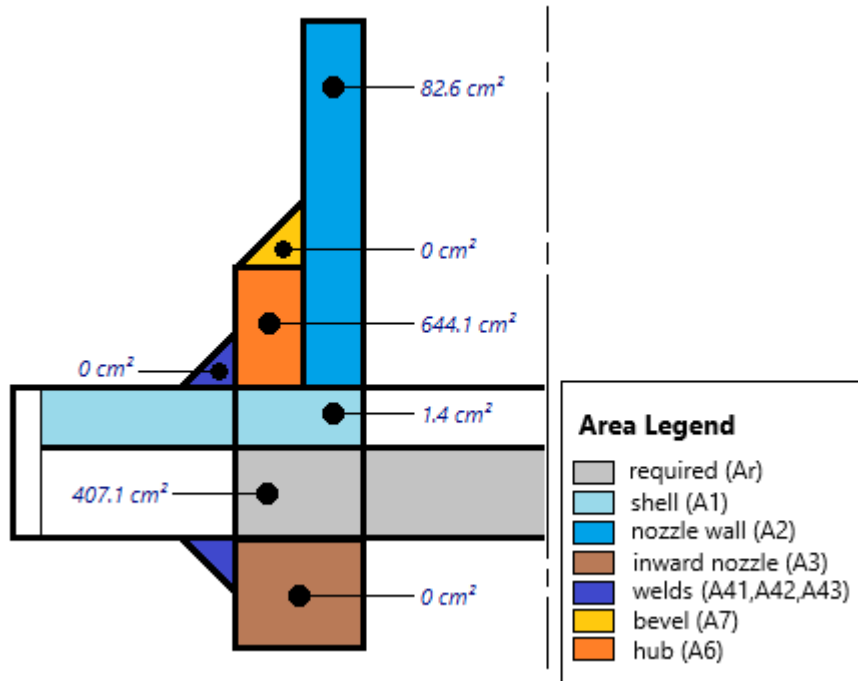
Nozl: 15 9:40am Apr 21,2026

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.99, 1)

= 0.986



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	407.079	9.692	NA
Area in Shell	A1	1.359	520.292	NA
Area in Nozzle Wall	A2	82.569	194.315	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	644.131	644.131	NA
TOTAL AREA AVAILABLE	Atot	728.059	1358.738	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (263•152•1+2•182•152•1•(1-0.99))

= 407.079 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)

= 365 (1 • 152 - 1 • 152) - 2 • 182

(1 • 152 - 1 • 152) • (1 - 0.99)

= 1.359 cm²



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Nozzle Calcs.: N1- inlet

Noz1: 15 9:40am Apr 21,2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 339) (30.3 - 18) 0.99$$

$$= 82.569 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_{r2} + (W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot f_{r2}$$

$$= (20^2 - 4) \cdot 0.99 + (0^2 - 0) \cdot 0.99$$

$$= 0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(339, 500, 150)) \cdot (185 - 33.3) \cdot 0.99$$

$$= 644.131 \text{ cm}^2 \text{ Includes Bevel Area of } 194.737 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 20.9706 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 5.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 154.6220 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 154.6220 \text{ mm}$
Wall Thickness, shell/head, external pressure	$tr_{b2} = 3.8284 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(155, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(21, 11.3)$$

$$= 20.9706 \text{ mm}$$

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

$$= \max(t_{UG-45}, tr_{B31.3})$$

$$= \max(21, 15.5)$$

$$= 20.9706 \text{ mm}$$

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = $0.875 \cdot 33.325 = 29.159 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 52.4,	Allowable	: 136.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 290.1 N./mm ²	Passed
Occasional	: 33.4,	Allowable	: 180.9 N./mm ²	Passed
Shear	: 12.9,	Allowable	: 95.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 29.2$, $t_r = 17.7$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.68$, Temp. Reduction = $18 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	5 °C



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Nozzle Calcs.: N1- inlet

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Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -36 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 17163.82/25510.00 = 0.673

Weld Size Calculations, Description: N1- inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (407 - 1.36 + 2 \cdot 182 \cdot 0.99 \cdot (1 \cdot 152 - 152)) \cdot 138)$$

$$= 572824.12 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (727 + 0.34108\text{E-}05 - 0 \cdot 0.99) \cdot 138$$

$$= 1022581.06 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (727 + 0 + 0.34108\text{E-}05 + (546)) \cdot 138$$

$$= 1791223.62 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (727 + 0 + 0.34108\text{E-}05 + 0 + (546)) \cdot 138$$

$$= 1791223.62 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 628 \cdot 20 \cdot 0.49 \cdot 136$$

$$= 134032. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 223) \cdot (185 - 3) \cdot 0.7 \cdot 136$$

$$= 1238628. \text{ Kgf}$$



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Nozzle Calcs.: N1- inlet

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Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 628 \cdot (155 - 3) \cdot 0.74 \cdot 138$$

$$= 1560967. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (134032 + 1238628) = 1372659 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (134032 + 0 + 1560966 + 0) = 1694998 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (134032 + 1560966 + 0) = 1694998 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1372659 Kgf, must exceed W = 572824 Kgf or W1 = 1022581 Kgf

Path 2-2 = 1694998 Kgf, must exceed W = 572824 Kgf or W2 = 1791223 Kgf

Path 3-3 = 1694998 Kgf, must exceed W = 572824 Kgf or W3 = 1791223 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 17183 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 44.6870 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 699.6870 mm

Input Echo, WRC297 Item 1, Description: N1- inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2250.000	mm
Shell Thickness	155.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	257.200	mm
Nozzle Thickness	250.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 136.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.



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Nozzle Calcs.: N1- inlet

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Design Internal Pressure	Dp	17163.82	KPa.
Radial Load	P	-3823.00	Kgf
Circumferential Shear	Vc	3058.00	Kgf
Longitudinal Shear	Vl	3058.00	Kgf
Circumferential Moment	Mc	2753.01	Kg-m.
Longitudinal Moment	Ml	2753.01	Kg-m.
Torsional Moment	Mt	3211.01	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, D_v/T_v (16.516) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_v (4.885) is < 5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_n (3.029) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 757.2 + 2 \cdot 1.65 \cdot \sqrt{(1204.0(155.0 - 3.0))}$$

$$= 2168.922 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2408.000 mm
Nozzle Outside Diameter	(d) =	757.200 mm
Vessel Thickness used	(T) =	152.000 mm
Nozzle Thickness used	(t) =	247.000 mm
T / t	=	0.615
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.252$$

Nr/P	=	0.097
Mr/P	=	0.187
M0/P	=	0.056
N0/P	=	0.120
MrD/Mc	=	0.296
NrDL/Mc	=	0.107
M0d/Mc	=	0.096
N0DL/Mc	=	0.048
MrD/Ml	=	0.168
NrDL/Ml	=	0.087
M0D/Ml	=	0.056
N0DL/Ml	=	0.130

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	119	136	119	136
Outplane Stress Summary	119	136	120	135



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Nozzle Calcs.: N1- inlet

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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	59	59	63	56

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	119	136	120	135

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59
Outplane Stress Summary	59	60	61	58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-2	2	2	-2
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	119	136	119	136
Inplane Stress Summary	118	137	124	132

Vessel Stresses (N./mm²):



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Nozzle Calcs.: N1- inlet

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CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	0	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			118	137	124	132

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)			0	0	0	0
Hoop Bending (P)			0	0	0	0
Hoop Membrane (Mc)			0	0	0	0
Hoop Bending (Mc)			0	0	0	0
Hoop Membrane (ML)			0	0	0	0
Hoop Bending (ML)			0	0	0	0
Hoop Pressure Stress			5	26	5	26
Hoop Stress Summary			5	26	6	26

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)			0	0	0	0
Axial Bending (P)			0	0	0	0
Axial Membrane (Mc)			0	0	0	0
Axial Bending (Mc)			0	0	0	0
Axial Membrane (ML)			0	0	0	0
Axial Bending (ML)			0	0	0	0
Axial Pressure Stress			2	2	2	2
Axial Stress Summary			2	2	4	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)			Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)			0	0	0	0
Shear due to (Vl)			0	0	0	0
Shear due to Torsion			0	0	0	0
Shear Stress Summary			0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress			5	26	6	26

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
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Nozzle Calcs.: N1- inlet

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(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	26	5	26
Hoop Stress Summary	5	26	5	26

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	1	2	4	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	5	26	5	26

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	119	136	119	136	119	136	119	136
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	-1	1	3	-3
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	2	-2	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	119.0	136.0	119.0	136.0	119.0	136.0	119.0	136.0



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Nozzle Calcs.: N1- inlet

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Pm+Pl (SUS)	119.0	136.0	119.0	136.0	119.0	136.0	119.0	136.0
Pm+Pl+Q (Total)	119.0	136.0	119.0	136.0	118.0	137.0	122.0	133.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	136.00	207.00	Passed
Pm+Pl+Q (TOTAL)	137.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	26	5	26	5	26	5	26
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	5.0	26.0	5.0	26.0	5.0	26.0	5.0	26.0
Pm+Pl (SUS)	5.0	26.0	5.0	26.0	5.0	26.0	5.0	26.0
Pm+Pl+Q (Total)	5.0	26.0	5.0	26.0	5.0	26.0	5.0	26.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	136.00	Passed
Pm+Pl (SUS)	26.00	204.00	Passed
Pm+Pl+Q (TOTAL)	26.00	408.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell inlet-N3

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Input, Nozzle Desc: Shell inlet-N3

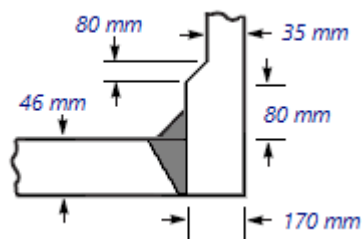
From: 40

Pressure for Reinforcement Calculations	P	4944.767	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2256.00	mm
Shell Finished (Minimum) Thickness	t	46.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4713.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	132.50	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	35.0000	mm
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		170.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	46.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: Shell inlet-N3



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Nozzle Calcs.: Shell inlet-N3 Nozl: 16 9:40am Apr 21,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 30.000 in.
Actual Thickness Used in Calculation 1.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4945 \cdot 1131 / (137 \cdot 1 - 0.6 \cdot 4945)$
= 41.7279 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4945 \cdot 381 / (133 \cdot 1 + 0.4 \cdot 4945)$
= 14.0101 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 135/0.5773
= 233.8270 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1396.0000	mm
Parallel to Vessel Wall, opening length	d	698.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	107.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 133/137)
= 0.967

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 133/137)
= 0.967

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.97, 1)
= 0.967



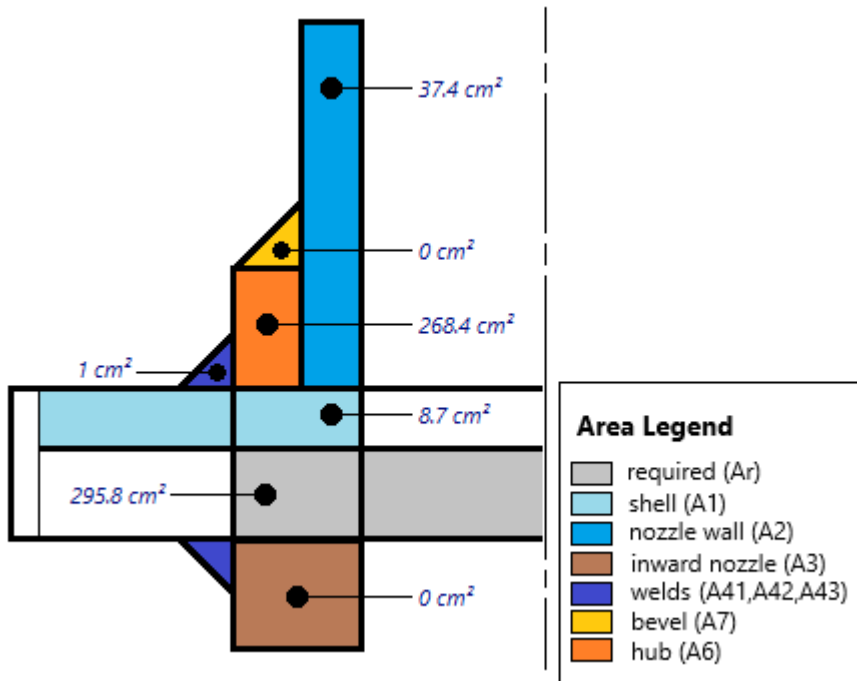
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Nozzle Calcs.: Shell inlet-N3

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	295.837	NA	NA
Area in Shell	A1	8.739	NA	NA
Area in Nozzle Wall	A2	37.408	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.967	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	268.379	NA	NA
TOTAL AREA AVAILABLE	Atot	315.494	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (698 \cdot 41.7 \cdot 1 + 2 \cdot 167 \cdot 41.7 \cdot 1 \cdot (1 - 0.97))$$

$$= 295.837 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 698(1 \cdot 43 - 1 \cdot 41.7) - 2 \cdot 167$$

$$(1 \cdot 43 - 1 \cdot 41.7) \cdot (1 - 0.97)$$

$$= 8.739 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 107)(32 - 14)0.97$$

$$= 37.408 \text{ cm}^2$$



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Nozzle Calcs.: Shell inlet-N3 Nozl: 16 9:40am Apr 21,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.97 + (0)^2 \cdot 0.97$$

$$= 0.967 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(107, 300, 80)) \cdot (170 - 35) \cdot 0.97$$

$$= 268.379 \text{ cm}^2 \text{ Includes Bevel Area of } 59.470 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.0101 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 44.7279 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 44.7279 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(44.7, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17, 11.3)$$

$$= 17.0101 \text{ mm}$$

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

$$= \max(tUG-45, trB31.3)$$

$$= \max(17, 12.4)$$

$$= 17.0101 \text{ mm}$$

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = 35.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 37.2,	Allowable	: 132.5 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 300.9 N./mm ²	Passed
Occasional	: 25.8,	Allowable	: 176.2 N./mm ²	Passed
Shear	: 7.2,	Allowable	: 92.8 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 46, tr = 41.4, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	13 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 13 °C

Weld Size Calculations, Description: Shell inlet-N3

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm



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Nozzle Calcs.: Shell inlet-N3

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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (296 - 8.74 + 2 \cdot 167 \cdot 0.97 \cdot (1 \cdot 43 - 41.7)) \cdot 137) \\
 &= 406806.41 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v \\
 &= (306 + 0.97 - 0 \cdot 0.97) \cdot 137 \\
 &= 428526.25 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v \\
 &= (306 + 0 + 0.97 + (139)) \cdot 137 \\
 &= 622572.44 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\
 &= (306 + 0 + 0.97 + 0 + (139)) \cdot 137 \\
 &= 622572.44 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 762 \cdot 10 \cdot 0.49 \cdot 133 \\
 &= 79243. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 365) \cdot (170 - 3) \cdot 0.7 \cdot 133 \\
 &= 1811120. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 762 \cdot (46 - 3) \cdot 0.74 \cdot 137 \\
 &= 532062. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (79243 + 1811120) = 1890363 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (79243 + 0 + 532062 + 0) = 611305 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (79243 + 532062 + 0) = 611305 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:



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Nozzle Calcs.: Shell inlet-N3 Nozl: 16 9:40am Apr 21,2026

Path 1-1 = 1890363 Kgf, must exceed W = 406806 Kgf or W1 = 428526 Kgf
 Path 2-2 = 611304 Kgf, must exceed W = 406806 Kgf or W2 = 622572 Kgf
 Path 3-3 = 611304 Kgf, must exceed W = 406806 Kgf or W3 = 622572 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5072 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 66.2924 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 412.2924 mm

Input Echo, WRC297 Item 1, Description: Shell inlet-N3:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2256.000	mm
Shell Thickness	46.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	1032.000	mm
Nozzle Thickness	170.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 132.50	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	4944.77	KPa.
Radial Load	P	-6881.02	Kgf
Circumferential Shear	Vc	4970.00	Kgf
Longitudinal Shear	Vl	4970.00	Kgf
Circumferential Moment	Mc	11468.00	Kg-m.
Longitudinal Moment	Ml	11468.00	Kg-m.
Torsional Moment	Mt	13188.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.071) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 1032.0 + 2 \cdot 1.65 \cdot \sqrt{(1152.5(46.0 - 3.0))}$$

$$= 1766.630 \text{ mm}$$



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Nozzle Calcs.: Shell inlet-N3

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WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2305.000 mm
 Nozzle Outside Diameter (d) = 1032.000 mm
 Vessel Thickness used (T) = 43.000 mm
 Nozzle Thickness used (t) = 167.000 mm
 T / t = 0.257
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.278$$

Nr/P = 0.056
 Mr/P = 0.098
 M0/P = 0.030
 N0/P = 0.064
 MrD/Mc = 0.262
 NrDL/Mc = 0.093
 M0d/Mc = 0.079
 N0DL/Mc = 0.086
 MrD/Ml = 0.080
 NrDL/Ml = 0.057
 M0D/Ml = 0.023
 N0DL/Ml = 0.074

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	127	132	127	132
Outplane Stress Summary	124	132	149	124

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	55	69	118	19

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1



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Nozzle Calcs.: Shell inlet-N3

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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	124	132	149	124
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-27	27	27	-27
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	39	82	105	36
-------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-92	92	92	-92
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	127	132	127	132

Inplane Stress Summary	52	200	249	25
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	53	200	249	37
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
(Stresses in the hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4



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Nozzle Calcs.: Shell inlet-N3 Nozl: 16 9:40am Apr 21,2026

Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	15	10	15
Hoop Stress Summary	8	13	16	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	3	4	8	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	8	13	16	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	15	10	15
Hoop Stress Summary	7	13	17	23

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	0	7	12	0



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Nozzle Calcs.: Shell inlet-N3

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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	7	13	17	23

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	127	132	127	132	127	132	127	132
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7
Circ. Q (SUS)	-2	2	14	-14	-71	71	113	-113
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-1	-1	5	5	-3	-3	7	7
Long. Q (SUS)	-7	7	49	-49	-21	21	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	127.0	132.0	127.0	132.0	127.0	132.0	127.0	132.0
Pm+Pl (SUS)	125.0	130.0	133.0	138.0	124.0	129.0	134.0	139.0
Pm+Pl+Q (Total)	123.0	132.0	147.0	124.0	53.1	200.0	247.0	37.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	132.00	137.00	Passed
Pm+Pl (SUS)	139.00	205.50	Passed
Pm+Pl+Q (TOTAL)	247.01	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	15	10	15	10	15	10	15
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7



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Nozzle Calcs.: Shell inlet-N3 Nozl: 16 9:40am Apr 21,2026

Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	10.0	15.0	10.0	15.0	10.0	15.0	10.0	15.0
Pm+Pl (SUS)	8.0	13.0	16.0	21.0	7.0	12.0	17.0	22.0
Pm+Pl+Q (Total)	8.0	13.0	16.0	21.0	8.0	12.0	17.0	23.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	132.50	Passed
Pm+Pl (SUS)	22.00	198.75	Passed
Pm+Pl+Q (TOTAL)	23.00	397.50	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

Input, Nozzle Desc: Shell Outlet- N3

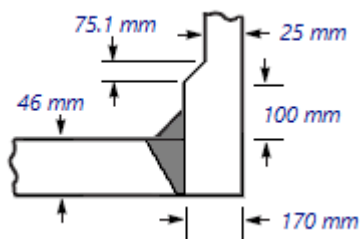
From: 40

Pressure for Reinforcement Calculations	P	4944.767	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2256.00	mm
Shell Finished (Minimum) Thickness	t	46.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4713.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	132.50	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	75.0555	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		170.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	46.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: Shell Outlet- N3



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 30.000 in.
Actual Thickness Used in Calculation 0.984 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4945 \cdot 1131 / (137 \cdot 1 - 0.6 \cdot 4945)$
= 41.7279 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4945 \cdot 381 / (133 \cdot 1 + 0.4 \cdot 4945)$
= 14.0101 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 145/0.5773
= 251.1474 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1436.0000	mm
Parallel to Vessel Wall, opening length	d	718.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	107.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 133/137)
= 0.967

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 133/137)
= 0.967

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.97, 1)
= 0.967



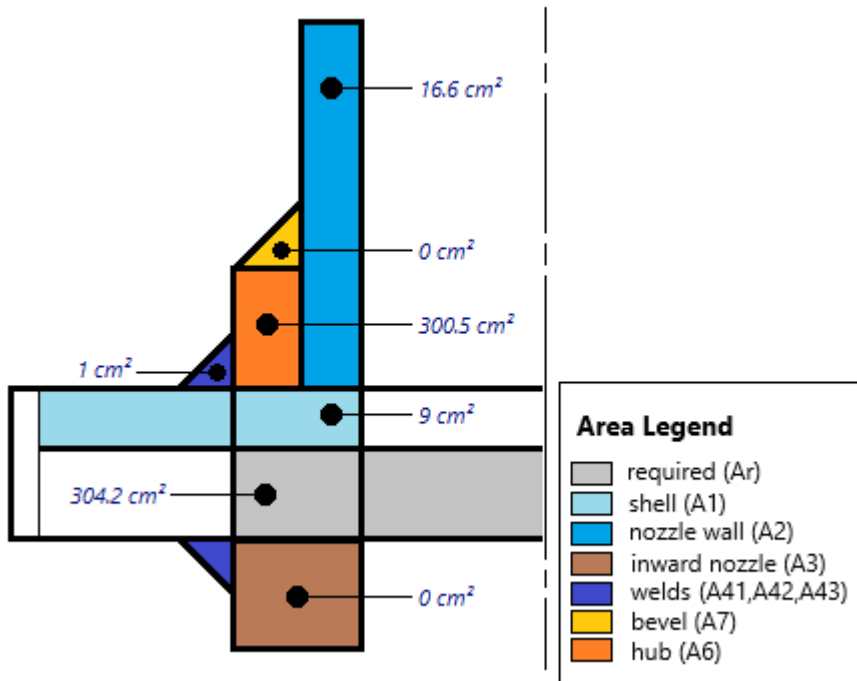
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Nozzle Calcs.: Shell Outlet- N

Nozl: 17 9:40am Apr 21,2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	304.182	NA	NA
Area in Shell	A1	8.994	NA	NA
Area in Nozzle Wall	A2	16.614	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.967	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	300.464	NA	NA
TOTAL AREA AVAILABLE	Atot	327.040	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (718 \cdot 41.7 \cdot 1 + 2 \cdot 167 \cdot 41.7 \cdot 1 \cdot (1 - 0.97))$$

$$= 304.182 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 718(1 \cdot 43 - 1 \cdot 41.7) - 2 \cdot 167$$

$$(1 \cdot 43 - 1 \cdot 41.7) \cdot (1 - 0.97)$$

$$= 8.994 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 107)(22 - 14)0.97$$

$$= 16.614 \text{ cm}^2$$



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.97 + (0)^2 \cdot 0.97$$

$$= 0.967 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(107, 300, 100)) \cdot (170 - 25) \cdot 0.97$$

$$= 300.464 \text{ cm}^2 \text{ Includes Bevel Area of } 19.985 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.0101 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 44.7279 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 44.7279 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(44.7, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17, 11.3)$$

$$= 17.0101 \text{ mm}$$

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

$$= \max(tUG-45, trB31.3)$$

$$= \max(17, 12.4)$$

$$= 17.0101 \text{ mm}$$

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = 25.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 55.1,	Allowable	: 132.5 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 283.0 N./mm ²	Passed
Occasional	: 39.1,	Allowable	: 176.2 N./mm ²	Passed
Shear	: 10.0,	Allowable	: 92.8 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 46, tr = 41.4, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	13 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 13 °C

Weld Size Calculations, Description: Shell Outlet- N

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (304 - 8.99 + 2 \cdot 167 \cdot 0.97 \cdot (1 \cdot 43 - 41.7)) \cdot 137)$$

$$= 418109.50 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (317 + 0.97 - 0 \cdot 0.97) \cdot 137$$

$$= 444300.53 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (317 + 0 + 0.97 + (139)) \cdot 137$$

$$= 638346.69 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (317 + 0 + 0.97 + 0 + (139)) \cdot 137$$

$$= 638346.69 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 762 \cdot 10 \cdot 0.49 \cdot 133$$

$$= 79243. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 370) \cdot (170 - 3) \cdot 0.7 \cdot 133$$

$$= 1835930. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 762 \cdot (46 - 3) \cdot 0.74 \cdot 137$$

$$= 532062. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (79243 + 1835930) = 1915173 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (79243 + 0 + 532062 + 0) = 611305 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (79243 + 532062 + 0) = 611305 \text{ Kgf}$$

Summary of Failure Path Calculations:



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

Path 1-1 = 1915173 Kgf, must exceed W = 418109 Kgf or W1 = 444300 Kgf
 Path 2-2 = 611304 Kgf, must exceed W = 418109 Kgf or W2 = 638346 Kgf
 Path 3-3 = 611304 Kgf, must exceed W = 418109 Kgf or W3 = 638346 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5072 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 66.2924 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 412.2924 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 71.09 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 718.00 mm

Input Echo, WRC297 Item 1, Description: Shell Outlet- :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2256.000	mm
Shell Thickness	46.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	1052.000	mm
Nozzle Thickness	170.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 132.50	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	4944.77	KPa.
Radial Load	P	-6879.98	Kgf
Circumferential Shear	Vc	4970.00	Kgf
Longitudinal Shear	Vl	4970.00	Kgf
Circumferential Moment	Mc	11468.00	Kg-m.
Longitudinal Moment	Ml	11468.00	Kg-m.
Torsional Moment	Mt	13188.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.188) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1052.0 + 2 \cdot 1.65 \cdot \sqrt{1152.5(46.0 - 3.0)}$$

$$= 1786.630 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2305.000 mm
 Nozzle Outside Diameter (d) = 1052.000 mm
 Vessel Thickness used (T) = 43.000 mm
 Nozzle Thickness used (t) = 167.000 mm
 T / t = 0.257
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.342$$

Nr/P = 0.056
 Mr/P = 0.096
 MØ/P = 0.030
 NØ/P = 0.064
 MrD/Mc = 0.261
 NrDL/Mc = 0.092
 MØd/Mc = 0.078
 NØDL/Mc = 0.086
 MrD/Ml = 0.079
 NrDL/Ml = 0.056
 MØD/Ml = 0.023
 NØDL/Ml = 0.073

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	127	132	127	132
Outplane Stress Summary	124	132	148	124

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	56	69	117	20

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

Outplane Shear (Mt)	1	1	1	1
---------------------	---	---	---	---

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	124	132	148	124
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-27	27	27	-27
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	40	81	104	37
-------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-90	90	90	-90
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	127	132	127	132

Inplane Stress Summary	54	198	246	28
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	54	198	246	37
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	2	2	2	2
--------------------	---	---	---	---



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	16	10	16
Hoop Stress Summary	8	14	16	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	3	4	8	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	8	14	16	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	16	10	16
Hoop Stress Summary	7	13	17	23

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	0	8	12	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	7	13	17	23

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	127	132	127	132	127	132	127	132
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7
Circ. Q (SUS)	-1	1	13	-13	-69	69	111	-111
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-1	-1	5	5	-2	-2	6	6
Long. Q (SUS)	-6	6	48	-48	-21	21	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	127.0	132.0	127.0	132.0	127.0	132.0	127.0	132.0
Pm+Pl (SUS)	125.0	130.0	133.0	138.0	124.0	129.0	134.0	139.0
Pm+Pl+Q (Total)	124.0	131.0	146.0	125.0	55.1	198.0	245.0	36.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	132.00	137.00	Passed
Pm+Pl (SUS)	139.00	205.50	Passed
Pm+Pl+Q (TOTAL)	245.01	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int.	Stress Values at
------------------------	------------------



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Nozzle Calcs.: Shell Outlet- N Nozl: 17 9:40am Apr 21,2026

Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	16	10	16	10	16	10	16
Circ. Pl (SUS)	-2	-2	6	6	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-1	1	1	-1	-4	4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	10.0	16.0	10.0	16.0	10.0	16.0	10.0	16.0
Pm+Pl (SUS)	8.0	14.0	16.0	22.0	8.0	14.0	16.0	22.0
Pm+Pl+Q (Total)	8.0	14.0	16.0	22.0	8.0	14.0	16.0	22.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	132.50	Passed
Pm+Pl (SUS)	22.00	198.75	Passed
Pm+Pl+Q (TOTAL)	22.00	397.50	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N5

Nozl: 18 9:40am Apr 21,2026

Input, Nozzle Desc: N5

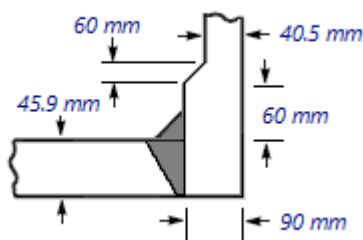
From: 50

Pressure for Reinforcement Calculations	P	4944.768	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	2256.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	45.9000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	132.50	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	45.9000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).





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Nozzle Calcs.: N5

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ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	12.812 in.
Actual Thickness Used in Calculation	1.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4945 \cdot 2262 \cdot 0.9 / (2 \cdot 137 \cdot 1 - 0.2 \cdot 4945)$
 $= 36.7817 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4945 \cdot 166 / (133 \cdot 1 - 0.6 \cdot 4945)$
 $= 6.3262 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 49.5 / 0.5773$
 $= 85.7580 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	662.8496	mm
Parallel to Vessel Wall, opening length	d	331.4248	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	107.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 133/137)$
 $= 0.967$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 133/137)$
 $= 0.967$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.97, 1)$
 $= 0.967$



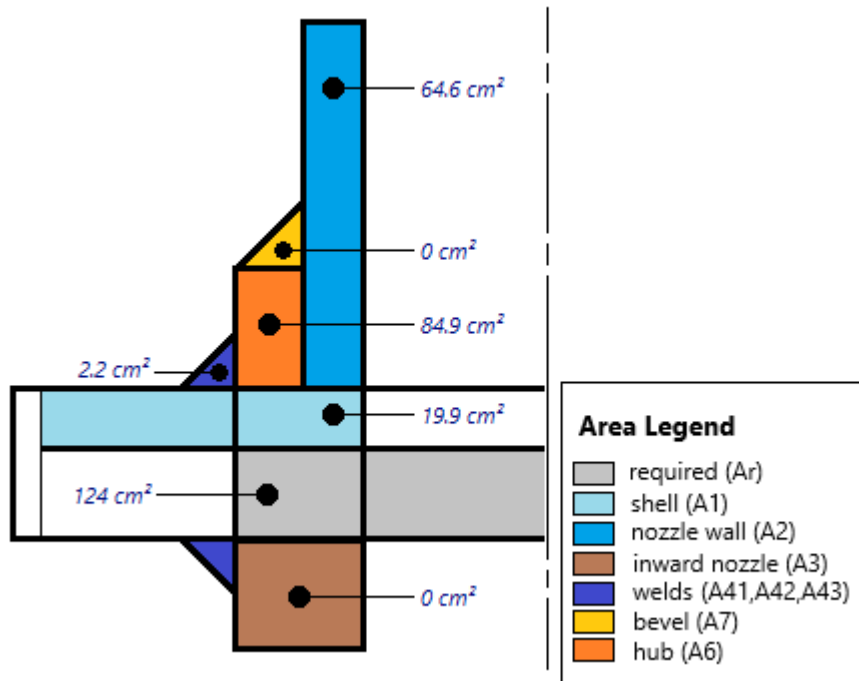
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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	124.005	NA	NA
Area in Shell	A1	19.928	NA	NA
Area in Nozzle Wall	A2	64.647	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.176	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	84.899	NA	NA
TOTAL AREA AVAILABLE	Atot	171.650	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (331 \cdot 36.8 \cdot 1 + 2 \cdot 87 \cdot 36.8 \cdot 1 \cdot (1 - 0.97))$$

$$= 124.005 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 331(1 \cdot 42.9 - 1 \cdot 36.8) - 2 \cdot 87$$

$$(1 \cdot 42.9 - 1 \cdot 36.8) \cdot (1 - 0.97)$$

$$= 19.928 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 107)(37.5 - 6.33)0.97$$

$$= 64.647 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 0.97 + (0)^2 \cdot 0.97$$

$$= 2.176 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(107, 500, 60)) \cdot (90 - 40.5) \cdot 0.97$$

$$= 84.899 \text{ cm}^2 \text{ Includes Bevel Area of } 27.435 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 9.3262 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 43.8278 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 43.8278 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(43.8, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(9.33, 11.3)$$

$$= 11.3312 \text{ mm}$$

Minimum wall thk. of nozzle necks including pressure, bending and shear loads [tBndUG-22]:

$$= \max(tUG-45, trB31.3)$$

$$= \max(11.3, 10.2)$$

$$= 11.3312 \text{ mm}$$

The value, trB31.3 was determined by iteration. This value is the required nozzle thickness considering sustained, occasional, expansion, pressure, bending and shear loads. The rules from ASME B31.3 have been applied.

Available Nozzle Neck Thickness = $0.875 \cdot 40.488 = 35.427 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 23.0,	Allowable	: 132.5 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 315.2 N./mm ²	Passed
Occasional	: 9.8,	Allowable	: 176.2 N./mm ²	Passed
Shear	: 7.6,	Allowable	: 92.8 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 45.9, tr = 36.5, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	6 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 6 °C

Weld Size Calculations, Description: N5

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm



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Nozzle Calcs.: N5

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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (124 - 19.9 + 2 \cdot 87 \cdot 0.97 \cdot (1 \cdot 42.9 - 36.8)) \cdot 137) \\
 &= 159775.92 \text{ Kg}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can} / 707)^2 \cdot f_r2) \cdot S_v \\
 &= (150 + 2.18 - 0 \cdot 0.97) \cdot 137 \\
 &= 211951.08 \text{ Kg}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v \\
 &= (150 + 0 + 2.18 + (72.2)) \cdot 137 \\
 &= 312805.91 \text{ Kg}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\
 &= (150 + 0 + 2.18 + 0 + (72.2)) \cdot 137 \\
 &= 312805.91 \text{ Kg}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 505 \cdot 15 \cdot 0.49 \cdot 133 \\
 &= 78841. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 209) \cdot (90 - 3) \cdot 0.7 \cdot 133 \\
 &= 540810. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 505 \cdot (45.9 - 3) \cdot 0.74 \cdot 137 \\
 &= 352089. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (78841 + 540810) = 619651 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (78841 + 0 + 352089 + 0) = 430930 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (78841 + 352089 + 0) = 430930 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:



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Path 1-1 = 619651 Kgf, must exceed W = 159775 Kgf or W1 = 211951 Kgf
 Path 2-2 = 430930 Kgf, must exceed W = 159775 Kgf or W2 = 312805 Kgf
 Path 3-3 = 430930 Kgf, must exceed W = 159775 Kgf or W3 = 312805 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5175 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 15.5056 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 561.4056 mm

Input Echo, WRC107/537 Item 1, Description: N5 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	4060.800	mm
Vessel Thickness	Tv	45.900	mm
Design Temperature	T1	275.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	137.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	325.425	mm
Nozzle Thickness	Tn	90.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	132.50	N./mm ²
Design Internal Pressure	Dp	4944.768	KPa.
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	3670.0	Kgf
Longitudinal Shear	(SUS)	(V1) V1	2651.0	Kgf
Circumferential Shear	(SUS)	(Vc) V2	2651.0	Kgf
Circumferential Moment	(SUS)	(Mc) M1	3262.0	Kg-m.
Longitudinal Moment	(SUS)	(M1) M2	3262.0	Kg-m.
Torsional Moment	(SUS)	Mt	3752.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



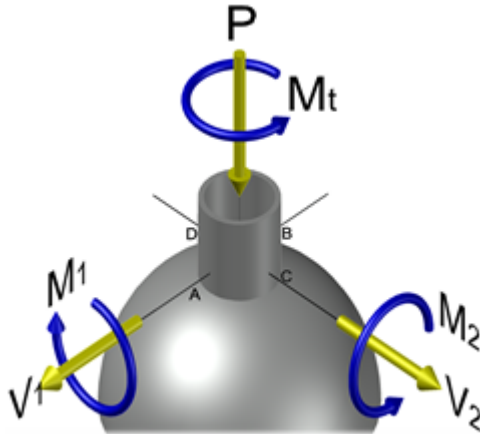
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WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 505.425 + 2 \cdot 1.65 \cdot \sqrt{(2054.85 (45.9 - 3.0))}$$

$$= 1485.214 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	3670.0	Kgf
Circumferential Shear	(VC) V2	2651.0	Kgf
Longitudinal Shear	(VL) V1	2651.0	Kgf
Circumferential Moment	(MC) M1	3262.0	Kg-m.
Longitudinal Moment	(ML) M2	3262.0	Kg-m.
Torsional Moment	MT	3752.0	Kg-m.

Dimensionless Param: U = 0.85 TAU = 5.00 (2.40) RHO = 0.49

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.04772	(A,B,C,D)
M(x) / P	SP 1	0.05189	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.06043	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.10322	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.06043	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.10322	(A,B,C,D)
N(y) * T / P	SP 1	0.07850	(A,B,C,D)
M(y) / P	SP 1	0.01553	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.07788	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.03502	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.07788	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.03502	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Membr. P	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9
Rad.	Bend. P	-6.1	6.1	-6.1	6.1	-6.1	6.1	-6.1	6.1
Rad.	Membr. MC	0.0	0.0	0.0	0.0	-3.5	-3.5	3.5	3.5



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Rad. Bend. MC	0.0	0.0	0.0	0.0	-36.3	36.3	36.3	-36.3
Rad. Memb. ML	-3.5	-3.5	3.5	3.5	0.0	0.0	0.0	0.0
Rad. Bend. ML	-36.3	36.3	36.3	-36.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-46.8	37.9	32.8	-27.6	-46.8	37.9	32.8	-27.6
Tang. Memb. P	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
Tang. Bend. P	-1.8	1.8	-1.8	1.8	-1.8	1.8	-1.8	1.8
Tang. Memb. MC	0.0	0.0	0.0	0.0	-4.6	-4.6	4.6	4.6
Tang. Bend. MC	0.0	0.0	0.0	0.0	-12.3	12.3	12.3	-12.3
Tang. Memb. ML	-4.6	-4.6	4.6	4.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-12.3	12.3	12.3	-12.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-20.2	8.0	13.5	-7.5	-20.2	8.0	13.5	-7.5
Shear VC	0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Tot. Shear	2.9	2.9	1.4	1.4	1.4	1.4	2.9	2.9
Str. Int.	47.1	38.2	32.9	27.7	46.9	37.9	33.2	28.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		117.2	117.2	117.2	117.2	117.2	117.2	117.2	117.2
Rad. Pl (SUS)		-4.5	-4.5	2.6	2.6	-4.5	-4.5	2.6	2.6
Rad. Q (SUS)		-42.3	42.3	30.2	-30.2	-42.3	42.3	30.2	-30.2
Long. Pm (SUS)		117.2	117.2	117.2	117.2	117.2	117.2	117.2	117.2
Long. Pl (SUS)		-6.1	-6.1	3.0	3.0	-6.1	-6.1	3.0	3.0
Long. Q (SUS)		-14.1	14.1	10.5	-10.5	-14.1	14.1	10.5	-10.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear Q (SUS)		2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Pm (SUS)		117.2	117.2	117.2	117.2	117.2	117.2	117.2	117.2
Pm+Pl (SUS)		113.1	113.1	120.8	120.8	113.1	113.1	120.8	120.8
Pm+Pl+Q (Total)		97.3	155.4	150.1	109.9	97.1	155.2	150.4	110.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.23	137.00	Passed
Pm+Pl (SUS)	120.83	205.50	Passed
Pm+Pl+Q (TOTAL)	155.38	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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ASME TS Calc:

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	4950.00 KPa.
Shell Maximum Operating Pressure	Psox,max	4215.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	46.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2256.000 mm
Shell Temperature for Internal Pressure	Ts	275.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	17000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	13537.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	155.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2250.000 mm
Channel Design Temperature	TEMPC	250.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4149
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	15.8800 mm
Total Straight Tube Length	Lt	5000.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	4660.00 mm
Design Temperature of the Tubes		250.00 °C
Tube Material		SA-213 TP316L
Tube Material UNS Number		S31603
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		103.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	114.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	22.2300 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	892.044 mm
Largest Center-to-Center Tube Distance	U1	50.0000 mm



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ASME TS Calc: Case: 1 9:40a Apr 21, 2026

Length of Expanded Portion of Tube	ltx	100.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	24327.8 cm ²
Perimeter of Tube Layout	Cp	7620.6 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	250.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	136.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	340.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	10.0000 mm
Tubesheet Outside Diameter	A	2680.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	2325.00 mm
Step 1 Depth on the Shellside	hs1	9.00 mm
Step 2 Diameter on the Shellside	ds2	2256.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2680.00 mm

Note: Tubesheet diameter is now: 2680.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	3375.0 cm ²
---------------------------	----	------------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2680.000 mm
Flange Inside Diameter	B	2256.000 mm
Flange Face Outside Diameter	Fod	2325.000 mm
Flange Face Inside Diameter	Fid	2256.000 mm
Gasket Outside Diameter	Go	2314.000 mm
Gasket Inside Diameter	Gi	2288.000 mm
Small end Hub thk.	g0	46.0000 mm
Large end Hub thk.	g1	68.0000 mm
Gasket Factor,	m	4.25
Gasket Design Seating Stress	y	69.64 N./mm ²
Flange Facing Sketch	Code	Sketch 1a
Gasket Thickness	tg	3.0000 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2550.000 mm
Nominal Bolt Diameter	dB	57.1500 mm
Thread Series	TEMA	Thread Series
Number of Bolts	n	64
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm



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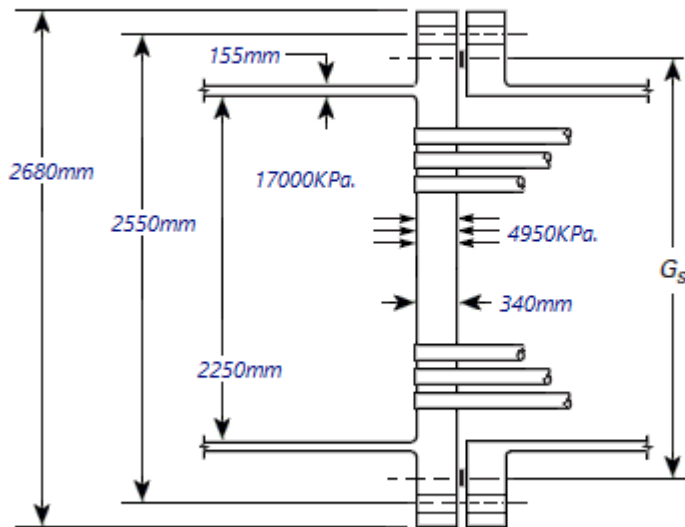
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ASME TS Calc:

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Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 324.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	250.0 °C	0.18759E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesheet Design Temp.	250.0 °C	0.17919E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	250.0 °C	0.18897E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (17000 \cdot (15.9/2 - 0)) / (103 \cdot 1 + 0.4 \cdot 17000)$
 $= 1.2294 + 0.0000 = 1.2294 \text{ mm}$



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ASME TS Calc:

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Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-4 at 250.00 °C
Elastic Modulus for Material 176649072.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	15.88	5552.04	9.62	50.0000	0.0118757	60.75

EMAWP = $(2.167 / (D/T) - 0.0833) * B$ = 8618 KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.0782	15.88	5552.04	14.73	50.0000	0.0050709	54.69

EMAWP = $(4 * B) / (3 * (D/T)) = (4 * 54.7) / (3 * 14.7) = 4951$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.2294 mm
Allowable Internal Pressure at Corroded thickness	23343.58 KPa.
Required Internal Design Pressure	17000.00 KPa.
Allowable External Pressure at Corroded thickness	8617.55 KPa.
Required External Design Pressure	4950.00 KPa.
Required Thickness due to Shell Side pressure	1.0782 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	2301.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * ro + dt$
 $= 2 * 892 + 15.9$
 $= 1799.968$ mm

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (22.2 - 15.9) / 22.2$
 $= 0.286$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Gs / Do$ (Configurations d, e, f)
 $= 2301 / 1800$
 $= 1.2784$

Ratio [pc]:

$= Dc / Do$ (Configurations a, e, f)
 $= 2250 / 1800$
 $= 1.2500$



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ASME TS Calc:

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ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 100/340$$

$$= 0.294 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(15.9 - 2 \cdot 1.65 (179191968/188968784) \cdot (103/136) \cdot (0.29); 15.9 - 2 \cdot 1.65)$$

$$= 15.1829 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 22.2 / \sqrt{ (1 - 4 \cdot \min(3375; 4 \cdot 1800 \cdot 22.2) / (\pi \cdot 1800^2)) }$$

$$= 22.9639 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (23 - 15.2) / 23$$

$$= 0.3388$$

$$h/p = 15.294647 ; \mu^* = 0.338835 ; E^* = 62808832 \text{ KPa.}$$

$$E^*/E = 0.332377 ; \nu^* = 0.331496$$

Note: As h/p (15.295) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2250 + 155) 155 }$$

$$= 0.0030 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.003 \cdot 187589856 \cdot 155^3 / (6(1 - 0.3^2))$$

$$= 380929015808.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2250 + 155 / \cos(0)) \cdot 380.93E+09 / 340^3 \cdot (1 + 340 \cdot 0.003 \cdot (1 + 1) / 2 + (340 \cdot 0.003)^2 / 2 \cdot 1)$$

$$= 353086208.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2250^2 / (4 \cdot 187589856 \cdot 155) \cdot (1 - 2250 \cdot 0.3 / (2(2250 + 155)))$$

$$= 0.37417E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) -$$

$$\zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$



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ASME TS Calc:

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$$\begin{aligned}
 &= 1.25 \cdot 380.93\text{E}+09 [0.037417024 (1 + 0.003 \cdot 1 \cdot 340) - \\
 &\quad 0.000 (1/340 + 1 \cdot 0.003/2)) \\
 &= 106753.5000 \text{ mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / D_o \\
 &= 2680/1800 \\
 &= 1.4889
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 &= (1 - v^*) / E^* \cdot (\lambda c + E \cdot \ln(K)) \\
 &= (1 - 0.33) / 62808832 \cdot (353086208 + 188968784 \cdot \ln(1.49)) \\
 &= 4.5587
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t] \\
 &= 1800^2/16 \cdot [(1.28 - 1) \cdot (1.28^2 + 1) \cdot 4950 - \\
 &\quad (1.25 - 1) \cdot (1.25^2 + 1) \cdot 17000] \\
 &= -1470267008.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]

$$\begin{aligned}
 &= M_{ts} + (\omega c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o) \\
 &= -1470267008 + (1814809472) + 2298549 \cdot (2550 - 2301) / (2 \cdot \pi \cdot 1800) \\
 &= 840495296.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M^*) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F) \\
 &= (840495296) - 1800^2/32 \cdot 4.56 (4950 - 17000) / (1 + 4.56) \\
 &= 1151742848.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2/64 \cdot (3 + v^*) \cdot (P_s - P_t) \\
 &= 1151742848 + 1800^2/64 \cdot (3 + 0.33) \cdot (4950 - 17000) \\
 &= -880501504.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| M_p | ; | M_o |) \\
 &= \max(| 1151742848 | ; | -880501504 |) \\
 &= 1151742848.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - h_g')^2) \\
 &= 6 \cdot 1151742848 / (0.34 (340 - 5)^2) \\
 &= 181.7413 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 136 \\
 &= 272.00 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - h_g')^2) \\
 &= 6 \cdot 1148025600 / (0.35 (276 - 5)^2) \\
 &= 271.8806 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$



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ASME TS Calc:

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$$= 276 + 0 + 0$$

$$= 275.6355 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS})$$

$$= \max(276; 124)$$

$$= 275.6355 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |4950 - 17000|$$

$$= 12050.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.29 \cdot 340 / 1800 \cdot \min(0.8 \cdot 136; 0.533 \cdot 204)$$

$$= 23.468 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.29 \cdot 109) \cdot (4 \cdot 24328 / 7621) \cdot |4950 - 17000| + 0$$

$$= 123.8604 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 155.000 mm

Min. Channel len. adjacent to the tubesheet is: 1062.991mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2250^2 / (4 \cdot 155 (2250 + 155) \cdot \cos(0)) \cdot 17000$$

$$= 57.721 \text{ N./mm}^2$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 85638200 / 155^2 \cdot [(0.037417024 \cdot 1 \cdot 0.003 - 0.000 \cdot 1) \cdot 17000 - 6((1 - 0.33) / 62808832) \cdot 1800 / 340^3 \cdot (1 + 0.003 \cdot 340 / 2 \cdot 1) \cdot (1151742848 + 1800^2 / 32 \cdot (4950 - 17000))]$$

$$= 208.804 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 266.524 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation (option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 187589856 \cdot \sqrt{(1.5 \cdot 138 / 267)}$$

$$= 165318784.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$



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ASME TS Calc:

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$$= 6 \cdot 1145784448 / (0.34 \cdot (340 - 5)^2)$$

$$= 180.8 \text{ N./mm}^2 \text{ (Should be } \leq 272.0 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:
= h + Cats + Catc = 276 + 0 + 0 = 276 mm

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:
= max(HreqB; HreqS) = max(276 ; 124) = 276 mm

Tube Metal Area [At]:
= $\pi (d_o - t) t$
= $\pi (15.9 - 1.65) 1.65$
= 0.738 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
= $\pi \cdot t (d_o - t) S$
= $\pi \cdot 1.65 (15.9 - 1.65) 103$
= 774.730 Kgf

Groove Weld Strength [Fg]:
= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
= $0.85 \cdot \pi \cdot 3 (15.9 + 0.67 \cdot 3) 103$
= 774.7297 Kgf

Weld Strength Factor [fw]:
= S / S_w where S_w = min(S and S_t)
= 103/103
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
= $0.55 \cdot \pi \cdot 3 (15.9 + 0.67 \cdot 3) 103$
= 774.7297 Kgf

Minimum Length of Weld [ar]:
= $2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$
= $2 \left(\sqrt{(0.75 \cdot 15.9)^2 + 1.07 \cdot 1.65 \cdot (15.9 - 1.65) 1 \cdot 1} - 0.75 \cdot 15.9 \right)$
= 2.023 mm

Minimum Required Fillet Weld Leg [afr]: 1.0117 mm
Minimum Required Groove Weld Leg [agr]: 1.0117 mm

Tube-Tubesheet Jt allowable, 775 is >= tube strength 775 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	180.8 <=	272.0 N./mm ²	Ok
Tubesheet shear stress	39.6 <=	108.7 N./mm ²	Ok
Stress in Channel at Tubesheet	266.5 <=	432.0 N./mm ²	Ok



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ASME TS Calc:

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Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	275.635	340.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.012	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.012	3.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

	--Reqd. Thk. + CA		----	Tubesheet	Stresses		Case	Pass/
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	327.301	132.057	255	272	56	109	Ps+Pt D1	Ok
D2uc	162.843	...	102	272	16	109	Ps+Pt D2	Ok
D3uc	275.635	...	181	272	40	109	Ps+Pt D3	Ok
D1c	338.107	132.057	269	272	58	109	Ps+Pt-c D1	Ok
D2c	170.803	...	105	272	17	109	Ps+Pt-c D2	Ok
D3c	282.341	...	186	272	41	109	Ps+Pt-c D3	Ok
Max:	338.1074	132.057	mm	0.988		0.534	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	...	397	432	Ok
D2uc	...	...	...	...	131	207	Ok
D3uc	...	...	...	...	267	432	Ok
D1c	...	...	...	...	421	432	Ok
D2c	...	...	...	...	139	207	Ok
D3c	...	...	...	...	281	432	Ok
Max	...	...	...	...	0.973		

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
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ASME TS Calc:

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Tubesheet thickness :	338.107	340.000	Ok
Tubesheet thickness flanged extension :	132.057	324.000	Ok
Tube thickness :	1.229	1.650	Ok
Tube-tubesheet fillet weld leg :	1.012	3.000	Ok
Tube-tubesheet groove weld leg :	1.012	3.000	Ok

Given Shell Thickness: 46.0000 mm

Given Channel Thickness: 155.0000 mm

Min Channel length of thk, (155.000) adj. to tubesheet: 1062.991 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18029.78	0.998	11850.49	1.000
Tubesheet Shear Stress	31812.82	1.000	31812.82	1.000
Tubesheet-Channel Junction	17144.01	0.998	8617.54	0.690
Tube Pressure Stress	23343.57	1.000	8617.54	1.000
Tubesheet Extension Stress	No Calc	No Calc	5338.22	...
Minimum MAWP	17144.01		5338.22	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18443.79	0.999	12515.55	1.000
Tubesheet Shear Stress	33584.27	1.000	33584.29	1.000
Tubesheet-Channel Junction	18443.79	0.999	12515.55	1.000
Tube Pressure Stress	26062.56	1.000	13848.93	1.000
Tubesheet Extension Stress	No Calc	No Calc	5338.22	...
Minimum MAPnc	18443.79		5338.22	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Channel Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
 = max(tubesheet thk/4, min(tubesheet thk, channel thickness))
 = max(340/4, min(340, 155))
 = 155.000 mm

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Shell Side MDMT calculation:

Governing thickness for the shell side:
 = tubesheet thickness/4
 = 340/4
 = 85.000 mm

Thickness Ratio = 0.93, Temperature Reduction per Fig. UCS 66.1 = 4 °C



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ASME TS Calc:

Case: 1 9:40a Apr 21, 2026

Min Metal Temp. w/o impact per UCS-66, Curve B	28 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	7 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(17000/18444, 4950/5338)

= 0.927

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Problems/Failures Summary:

Step: 32 9:40am Apr 21,2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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